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# THE USE OF FACES TO REPRESENT POINTS IN n-DIMENSIONAL SPACE GRAPHICALLY

BY

HERMAN CHERNOFF

TECHNICAL REPORT NO. 71

DECEMBER 27, 1971

PREPARED UNDER CONTRACT

N00014-67-A-0112-0051 (NR-042-993)

FOR THE OFFICE OF NAVAL RESEARCH

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| | |
|---|---|
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3. REPORT TITLE

THE USE OF FACES TO REPRESENT POINTS IN n-DIMENSIONAL SPACE GRAPHICALLY

4. DESCRIPTIVE NOTES (Type of report and inclusive dates)

TECHNICAL REPORT

5. AUTHOR(S) (Last name, first name, initial)

CHERNOFF, Herman

| 6. REPORT DATE | 7a. TOTAL NO. OF PAGES | 7b. NO. OF REFS |
|---|---|---|
| December 27, 1971 | 45 | 6 |

| 8a. CONTRACT OR GRANT NO. | 9a. ORIGINATOR'S REPORT NUMBER(S) |
|---|---|
| N00014-67-A-0112-0051 | #71 |
| b. PROJECT NO.<br>NR-042-993 | |
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13. ABSTRACT

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| 14. KEY WORDS | LINK A | | LINK B | | LINK C | |
|---|---|---|---|---|---|---|
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| cluster analysis<br>faces<br>graphical representation<br>multivariate data<br>profile | | | | | | |

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# THE USE OF FACES TO REPRESENT POINTS IN n-DIMENSIONAL SPACE GRAPHICALLY

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# THE USE OF FACES TO REPRESENT POINTS IN n-DIMENSIONAL SPACE GRAPHICALLY

by

Herman Chernoff

## 1. Introduction

Graphical representations serve to communicate essential information conveniently and effectively. They are also useful in exploratory work with data. In particular, a scatter diagram is a powerful device for indicating at a glance the essential relationships between two variables which have a bivariate distribution. In a field like cluster analysis, where the concept of cluster is not clear-cut, the actual data may have a profound effect on what the investigator would choose to call a cluster. In the bivariate case the scatter diagram can be used effectively to help decide which concept to use. For some problems the inadequacy of the classical linear techniques of normal multivariate analysis may be clearly revealed by graphs which can be used to suggest suitable transformations or other modifications of linear theory.

When dealing with multivariate data involving more than two variables, the scatter diagram can be used only in the limited form where two variables are studied at a time. This is unwieldy when the number of variables involved is large. Moreover, subtle relations or effects which require the simultaneous consideration of more than two variables may go undetected when this approach is used.

A new method of representing multivariate data graphically is described here. Briefly, it consists of representing a point in k-dimensional space by a picture of a face whose characteristics are determined by the position of the point. A sample of points in k-dimensional space is represented by a collection of faces.

In the next section, two illustrations are sketched briefly. In one of these where the investigator was interested in a cluster analysis, his task was merely to group together those faces which resemble each other. In the second, where the investigator was interested in detecting time points where a multivariate stochastic process changed character, he had to look at the sequence of faces corresponding to successive points in time to locate the places where the faces change character.

Following sections discuss the potential advantage of this graphical method over that of looking at numerical data and consider some alternative approaches to and predecessors of this method. Detailed documentation, including the data for the illustrative examples and the method of generating the faces, is contained in the appendix.

## 2. Illustrations

We present two examples illustrating this representation.

### Example 1. Fossil Data

Eight measurements were made on each of 88 nummulited specimens from the Eocene Yellow Limestone Formation of northwestern Jamaica. Two measurements thought to be age-dependent were discarded. One specimen (Number 34) was rejected because of a permutation in an early copy of the measurements for that specimen which cast doubt upon its

accuracy. The data and definition of the measurements appear in Table 2a of Appendix A4. The 87 faces corresponding to the 87 remaining specimens are presented in sequential order as indicated in Figure 1a. This order was selected after the data and had been grouped into three clusters.

The number at the bottom and left of each face is a randomly selected code number. Because the data were handled in two subgroups, these code numbers are repeated twice, but half were marked with a cross. The i.d. numbers were added for publication. It is immediately obvious how these faces divide into three distinct clusters. This division is obvious partly because of the special arrangement of the faces. When copies were made, separated and mixed up, and then given to people to cluster visually, these people selected the same clusters. On several occasions there were one or two discrepancies. The people, having the code number but not the sequence number, had no way of knowing except through the faces what grouping was expected.

A follow-up attempt to separate the large cluster of the first 40 faces (1-41 with 34 omitted) into subclusters seemed to be difficult, and the results of various individuals were inconsistent with one another. Since the ranges of the variables in the first 40 specimens were smaller than for the 87 specimens, it seemed reasonable to magnify the effects of variation by renormalizing the data according to the ranges in the first 40 specimens. A new set of faces was produced and is presented in Figure 1b. I clustered these visually. The groups were

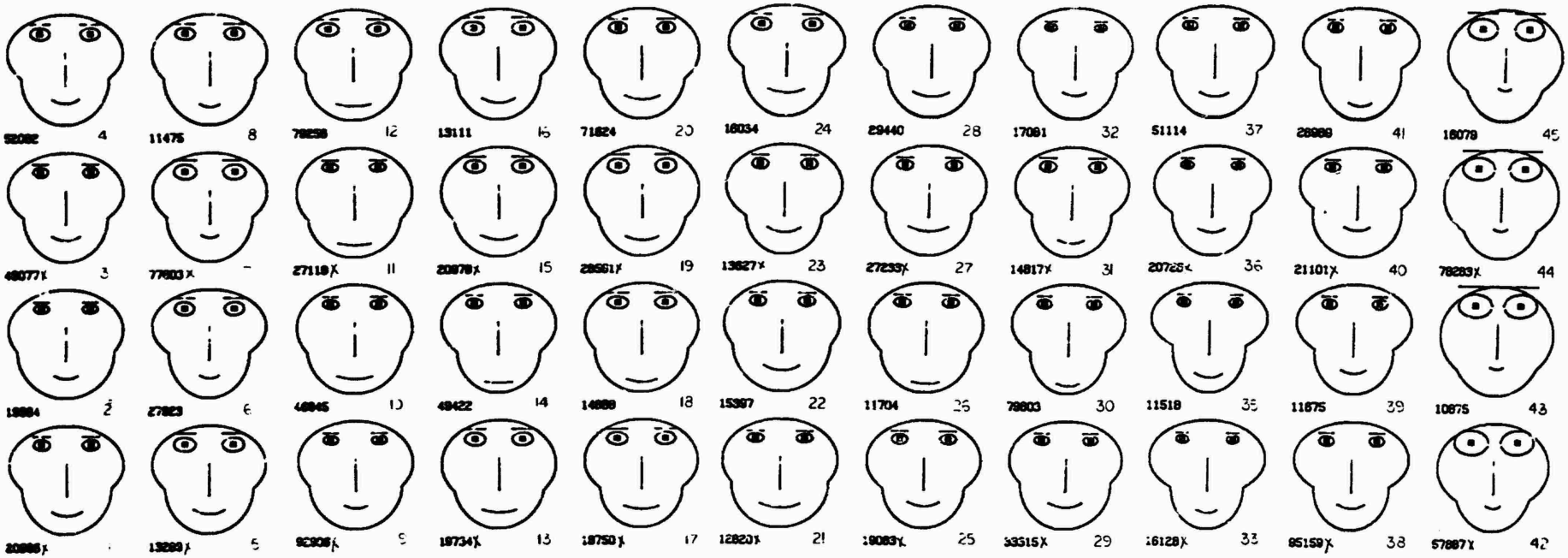
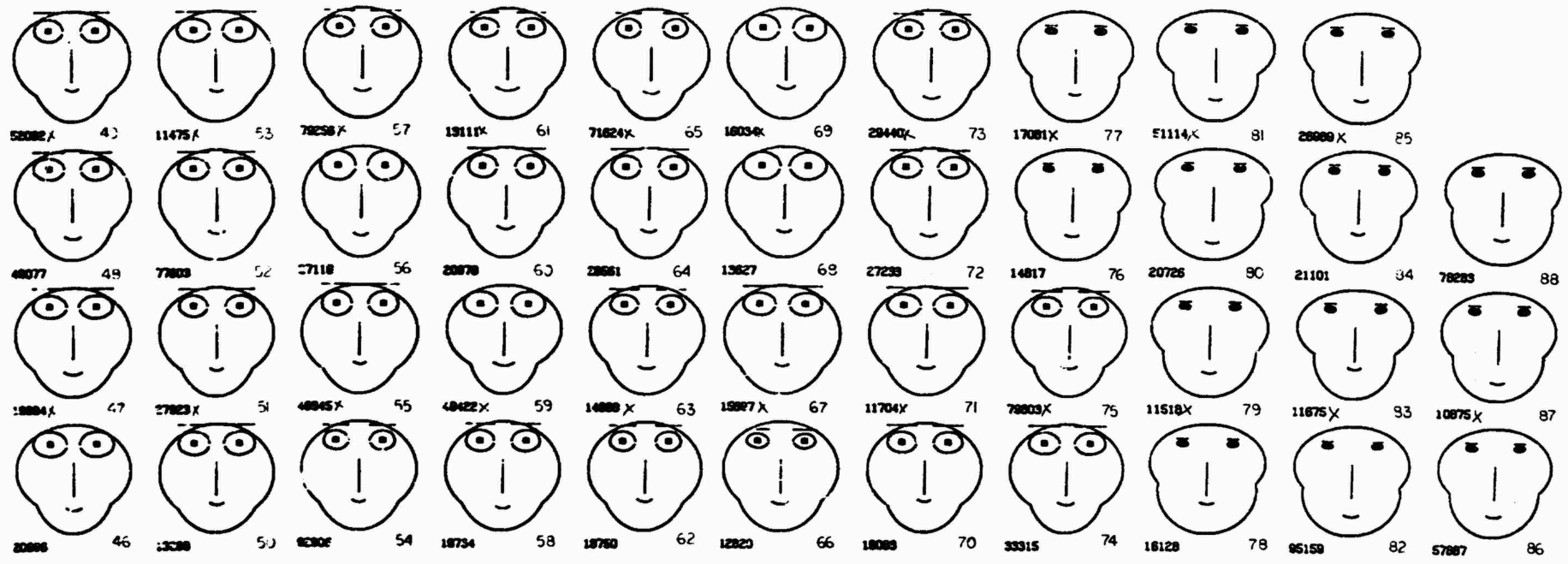


FIG. 1A



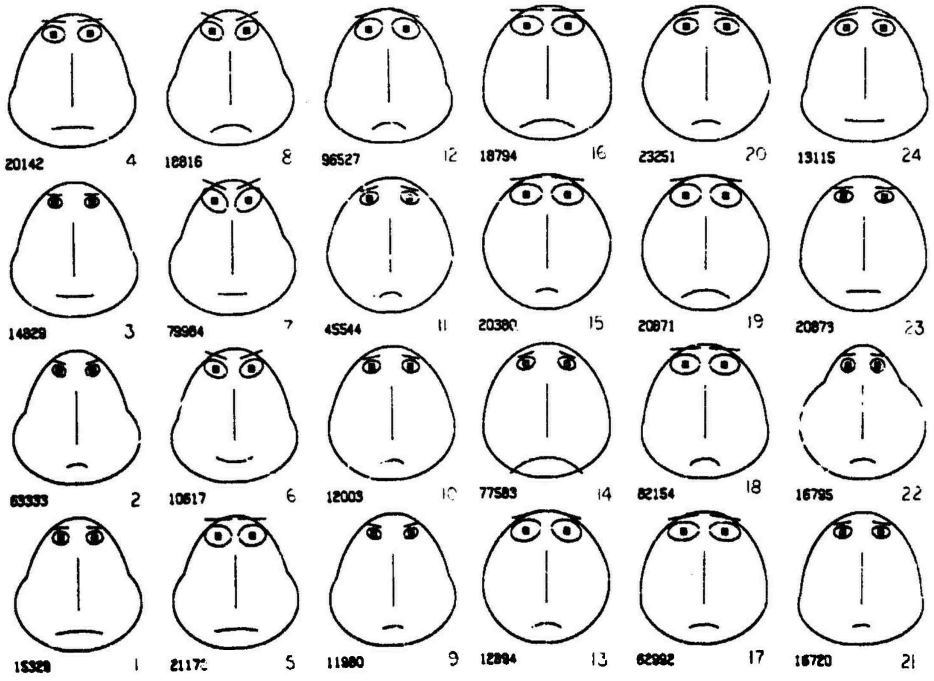

20142 4
18816 8
96527 12
18794 16
23251 20
13115 24
14829 3
78984 7
45544 11
20380 15
20871 19
20873 23
69333 2
10617 6
12003 10
77583 14
82154 18
16785 22
15328 1
21175 5
11980 9
12894 13
62992 17
16720 21

FIG. 1B

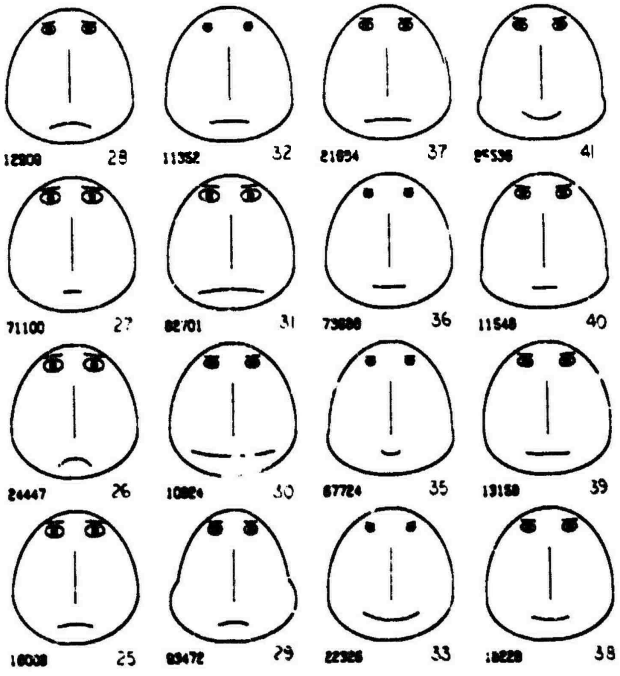
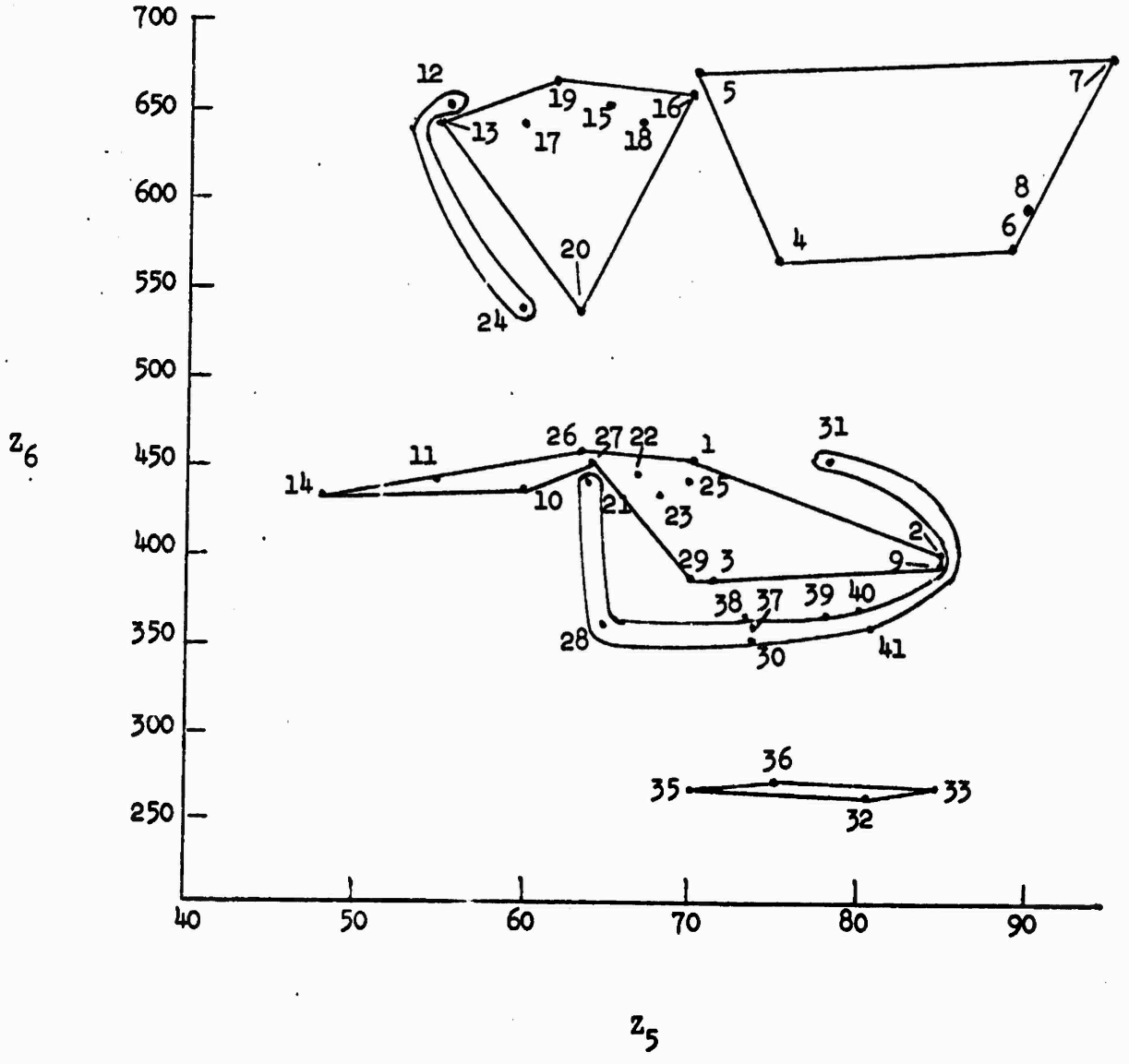

12908 28
11352 32
21834 37
95536 41
71100 27
82701 31
73698 36
11548 40
26447 26
10824 30
67724 35
13158 39
16038 25
83472 29
22388 33
18228 38

Figure 1C


700
650
600
550
500
450
400
350
300
250
40
50
60
70
80
90
$z_6$
$z_5$
12
19
15
16
13
17
18
5
7
8
6
4
20
24
11
26
27
22
1
31
14
10
21
23
25
2
29
3
9
38
37
39
40
28
30
41
36
35
33
32

I: (1,2,3,9,22,29)
II: (4,5,6,7,8)
III: (10,11,14,23,25,26,27)
IVa: (13,15,16,17,18,19,20)
IVb: (12,24)
V: (21,28,30,31,37,38,39,40,41)
VI: (32,33,35,36)

where IVb seemed to be similar to IVa but slightly different. Professor Switzer also clustered them visually. He obtained:

Ia: (1,2,3,9,22,29)
Ib: (4,5,12,24)
Ic: (13,15,16,17,18,19)
II: (6,7,8)
III: (10,11,14,20,21,23,25,26,27,28,31)
IV: (30,32,33,35,36,37,38,39,40,41)

which, though not in complete agreement with my groups, has substantial similarity. These grouping attempts were more ambitious than is ordinarily necessary for one can easily choose to leave peculiar cases out of the groupings. Furthermore these lists in numerical order do not indicate which specimens were obviously members of a group and which were regarded as borderline.

Finally, in connection with this example, a graph of $(Z_5,Z_6)$ for these specimens is presented since these variables seemed important in the set of 87 specimens. See Figure 1C.

Example 2. Geological Data

Mineral analysis data from a 4,500-foot core drilled from a Colorado mountainside yielded 12 variables. These represent assays of 7 mineral

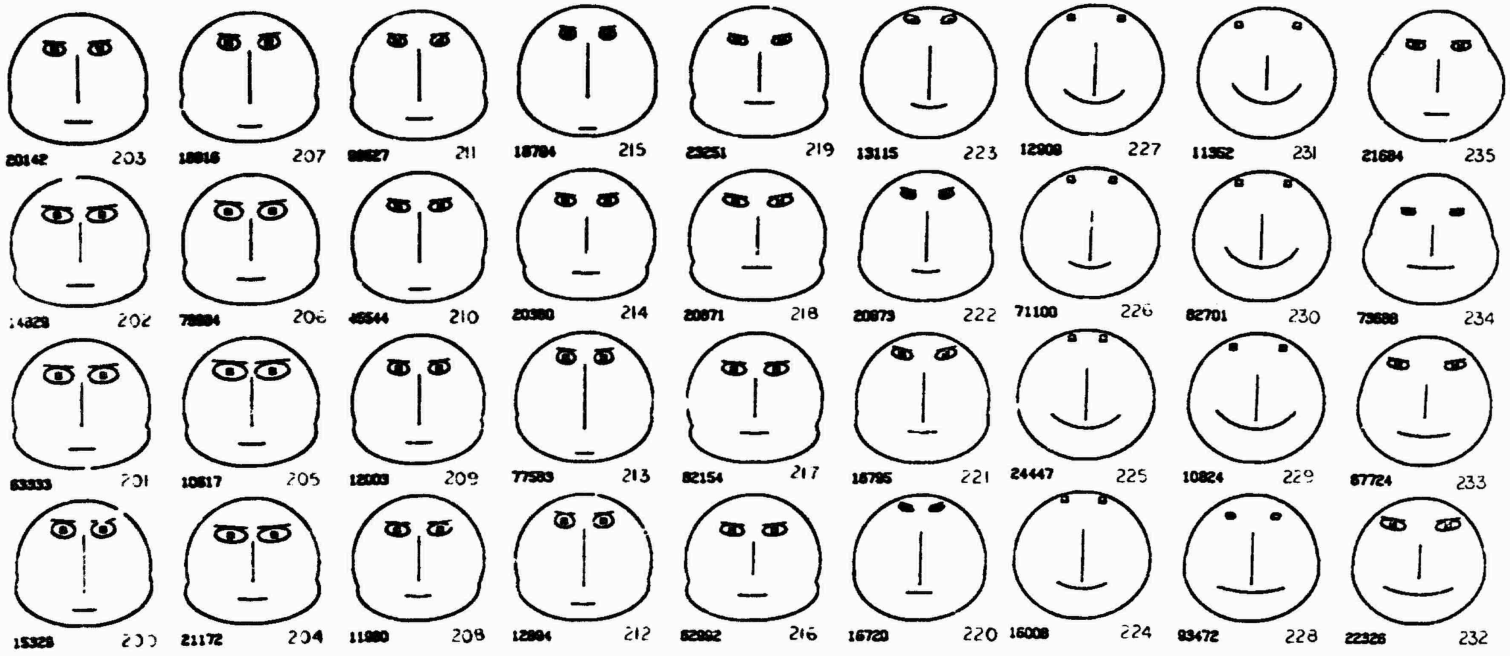
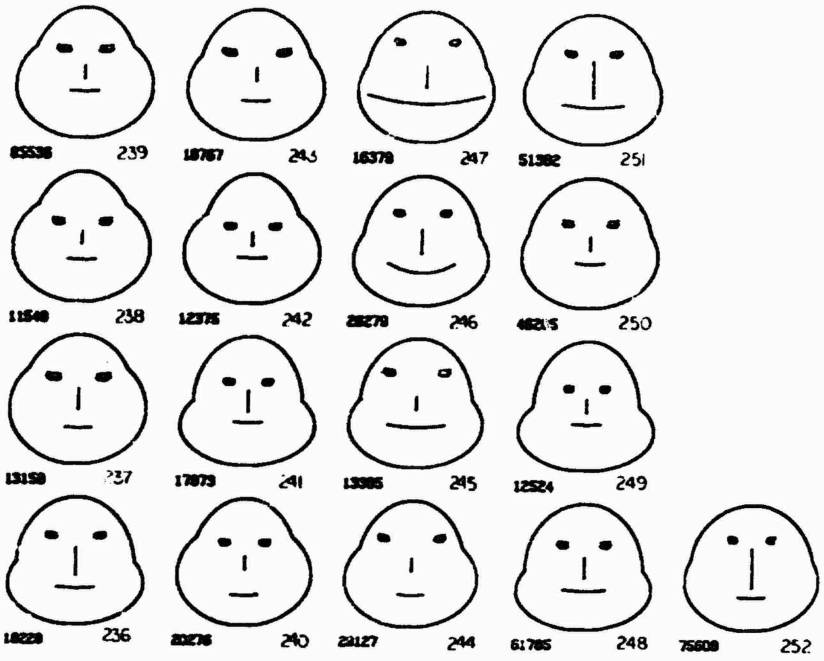

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18784 215
23251 219
13115 223
12908 227
11352 231
21684 235
14829 202
78804 206
45544 210
20380 214
20871 218
20873 222
71100 226
82701 230
79688 234
83333 201
10817 205
12003 209
77583 213
82154 217
16795 221
24447 225
10824 229
87724 233
15328 200
21172 204
11980 208
12884 212
82992 216
16720 220
16008 224
93472 228
22326 232

FIG. 2A


85536 239
18767 243
16378 247
51382 251
11540 238
12375 242
28278 246
48215 250
13158 237
17873 241
13985 245
12524 249
18228 236
20276 240
28127 244
61785 248
75608 252

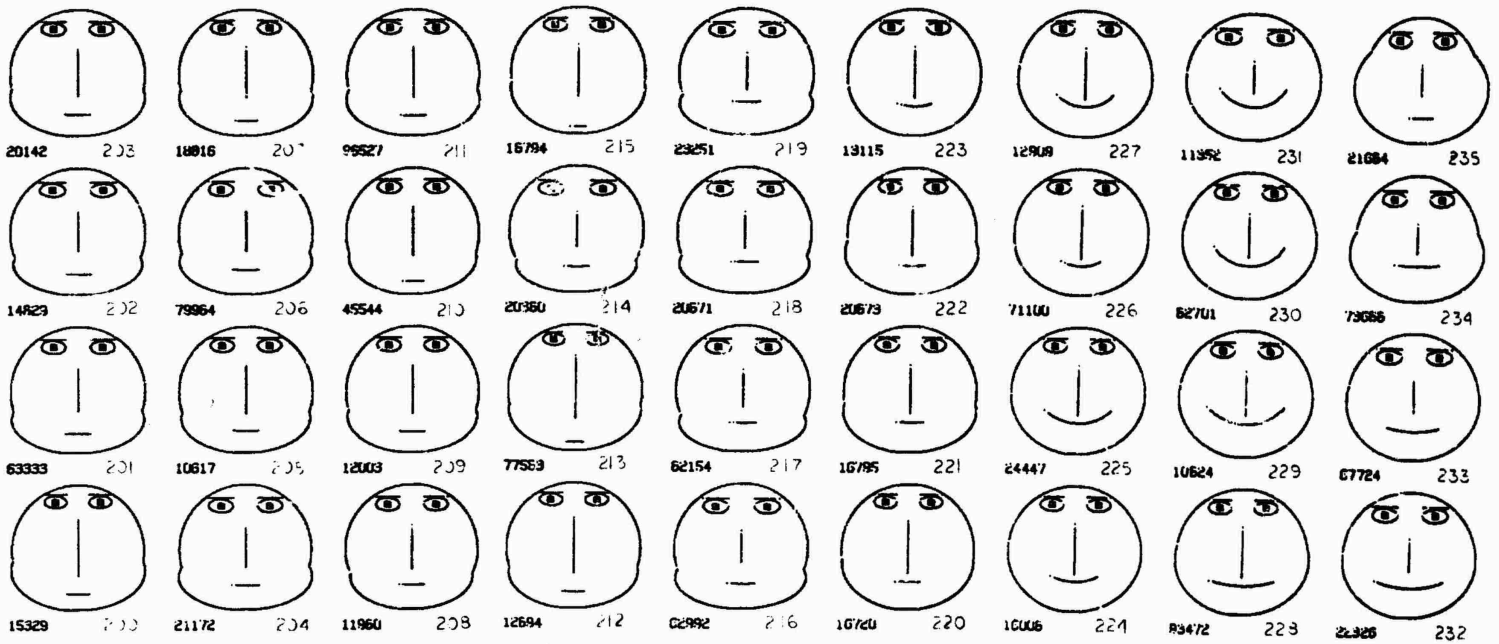
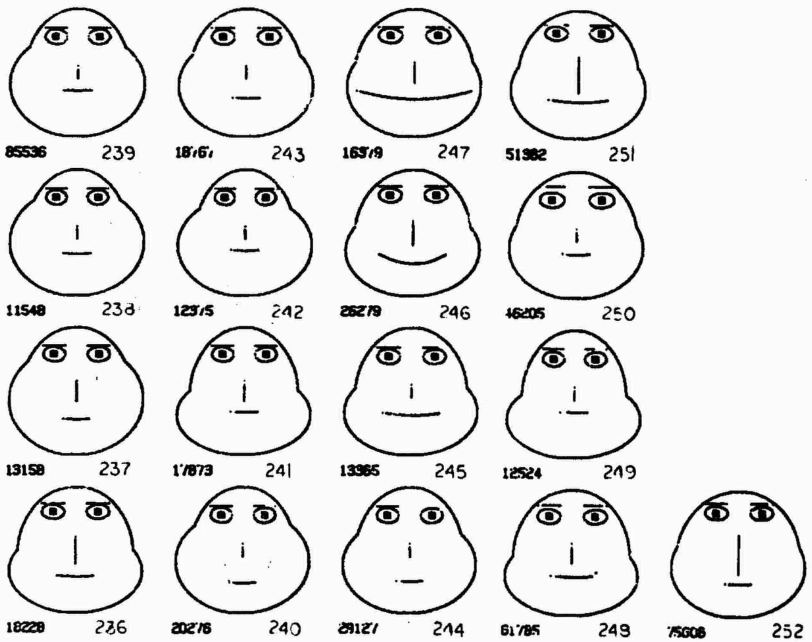

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16794 215
23251 219
13115 223
12908 227
11952 231
21684 235
14823 202
79964 206
45544 210
20360 214
20671 218
20679 222
71100 226
62701 230
73088 234
63333 201
10617 205
12003 209
77563 213
62154 217
16785 221
24447 225
10624 229
67724 233
15329 200
21172 204
11960 208
12694 212
62992 216
16720 220
10006 224
83472 228
22326 232

FIG. 2B


85536 239
18767 243
16978 247
51382 251
11548 238
12375 242
26279 246
46205 250
13158 237
17873 241
13365 245
12524 249
18228 236
20276 240
29127 244
61785 248
75008 252

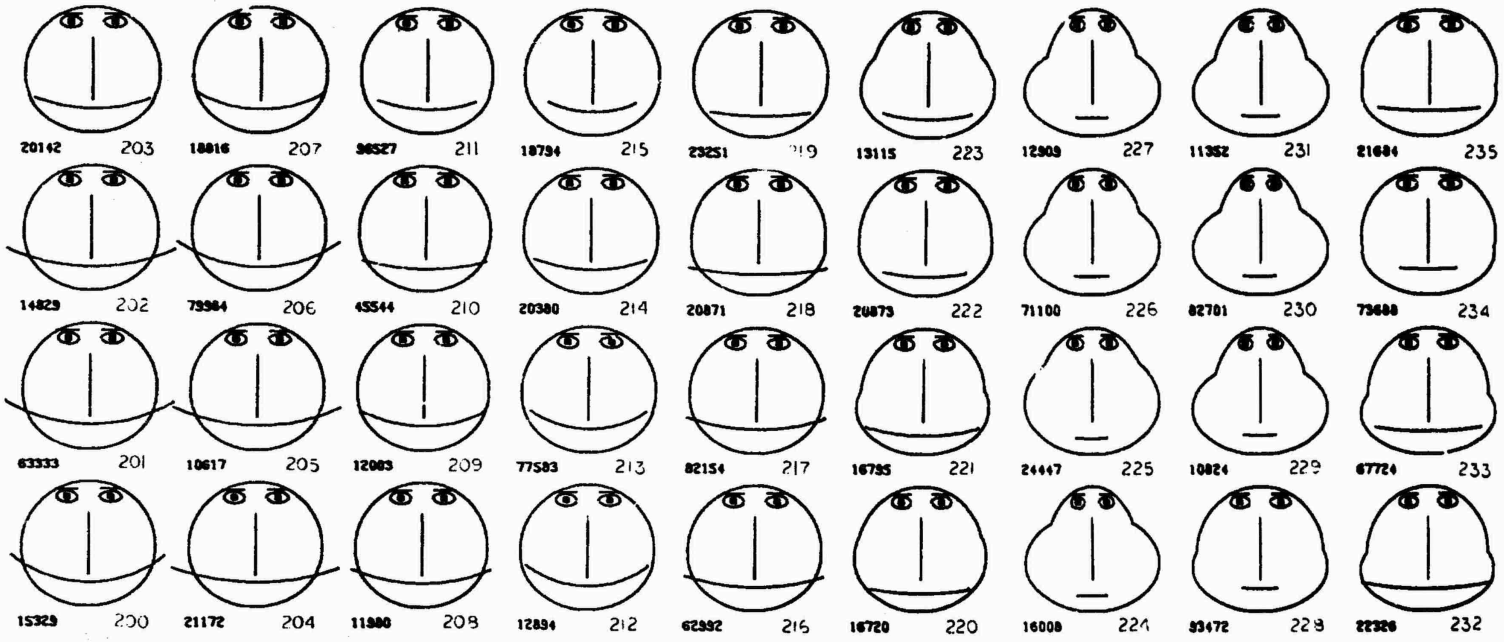
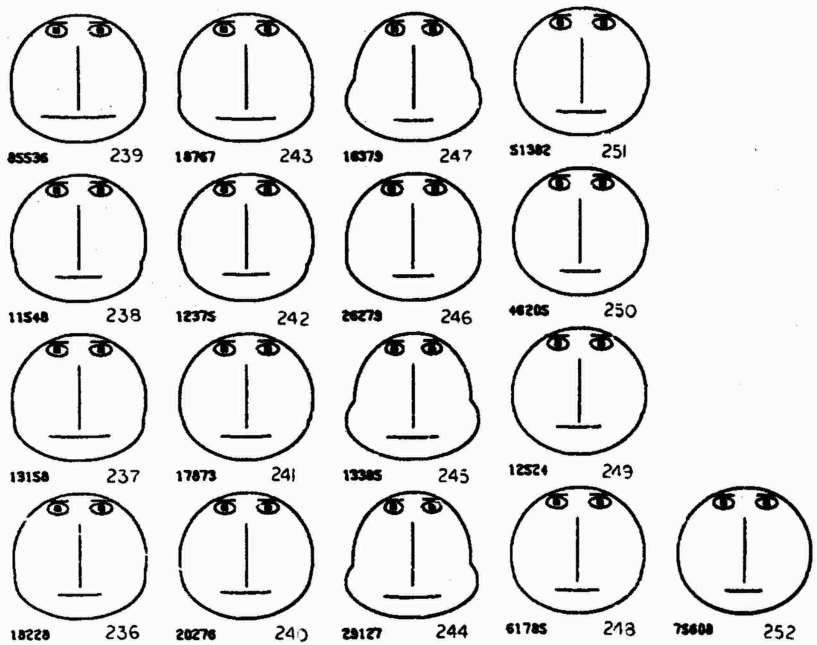

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38527 211
18794 215
23251 219
13115 223
12909 227
11352 231
21684 235
14829 202
73984 206
45544 210
20380 214
20871 218
20873 222
71100 226
82701 230
73688 234
63333 201
10617 205
12083 209
77583 213
82154 217
16795 221
24447 225
10824 229
67724 233
15329 200
21172 204
11880 208
12894 212
62392 216
16720 220
16008 224
93472 228
22326 232

FIG. 2C


85536 239
18767 243
16379 247
51382 251
11548 238
12375 242
26273 246
46205 250
13158 237
17873 241
13385 245
12524 249
18228 236
20276 240
29127 244
61785 248
75688 252

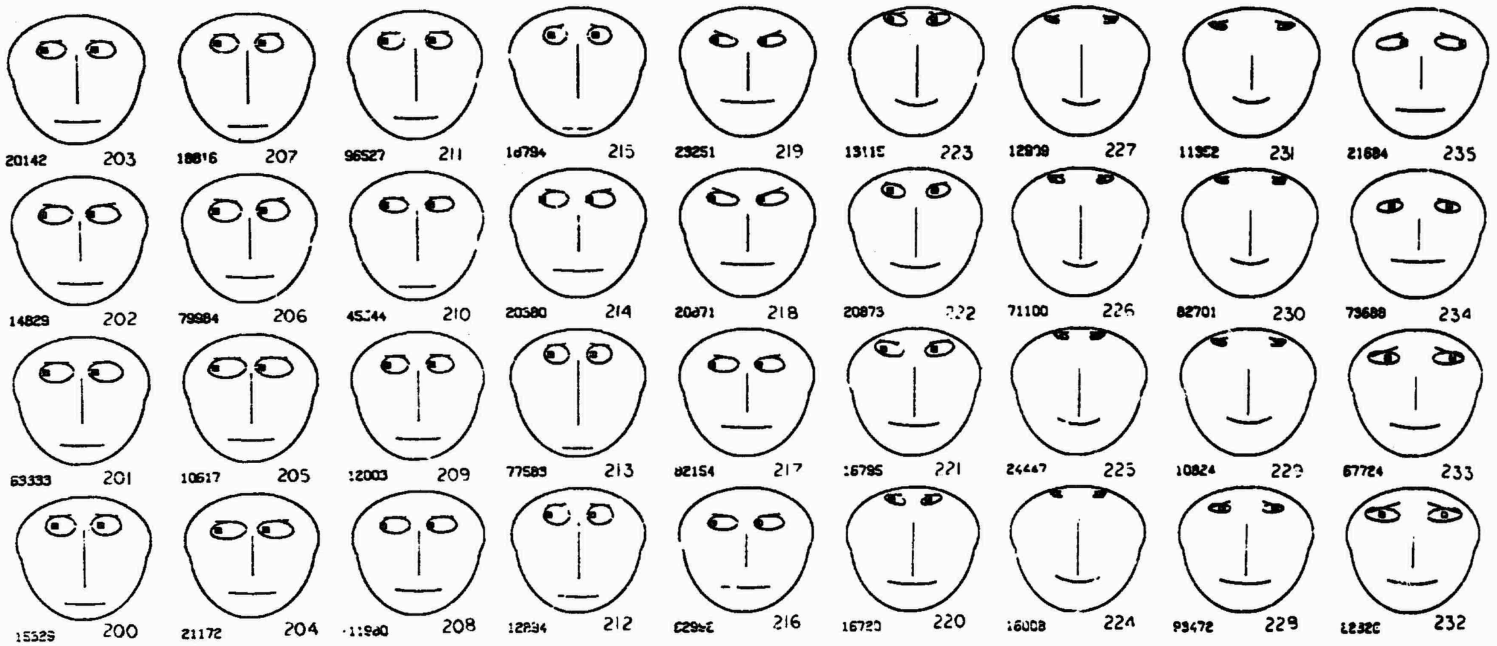
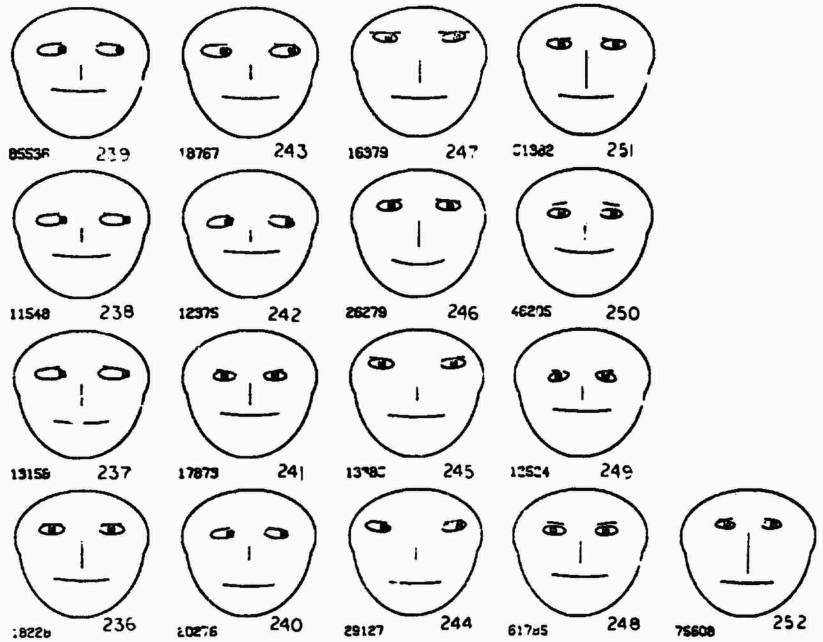

20142 203
18816 207
96527 211
16794 215
23251 219
13115 223
12919 227
11352 231
21684 235
14829 202
79984 206
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20871 218
20873 222
71100 226
82701 230
79688 234
63333 201
10617 205
12003 209
77583 213
82154 217
16795 221
24447 225
10824 229
67724 233
15525 200
21172 204
12894 212
16720 220
16008 224
93472 229
12320 232

FIG. 2D


85536 239
18767 243
16379 247
31382 251
11548 238
12375 242
26279 246
46275 250
13156 237
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13780 245
12524 249
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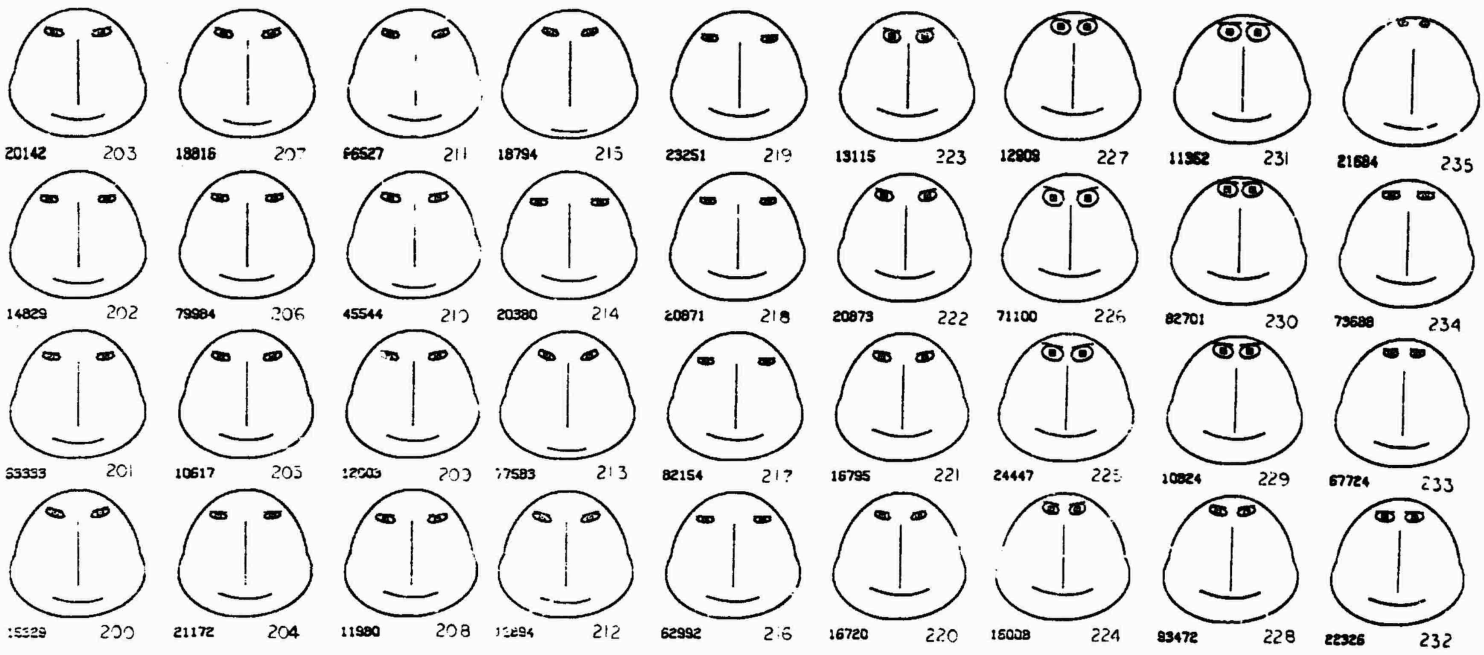
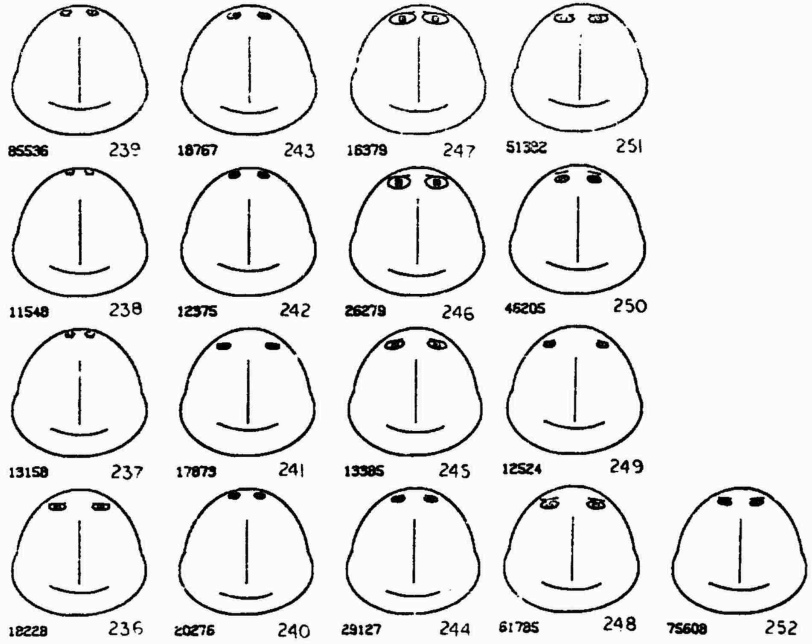

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18816 207
66527 211
18794 215
23251 219
13115 223
12909 227
11362 231
21684 235
14829 202
79984 206
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20380 214
20871 218
20873 222
71100 226
82701 230
79688 234
63333 201
10617 205
12003 209
77583 213
82154 217
16795 221
24447 225
10824 229
67724 233
15329 200
21172 204
11980 208
12894 212
62992 216
16720 220
16008 224
83472 228
22326 232

FIG. 2E


85536 239
18767 243
16379 247
51382 251
11548 238
12375 242
26279 246
46205 250
13158 237
17873 241
13385 245
12524 249
18228 236
20276 240
29127 244
61785 248
75608 252

12

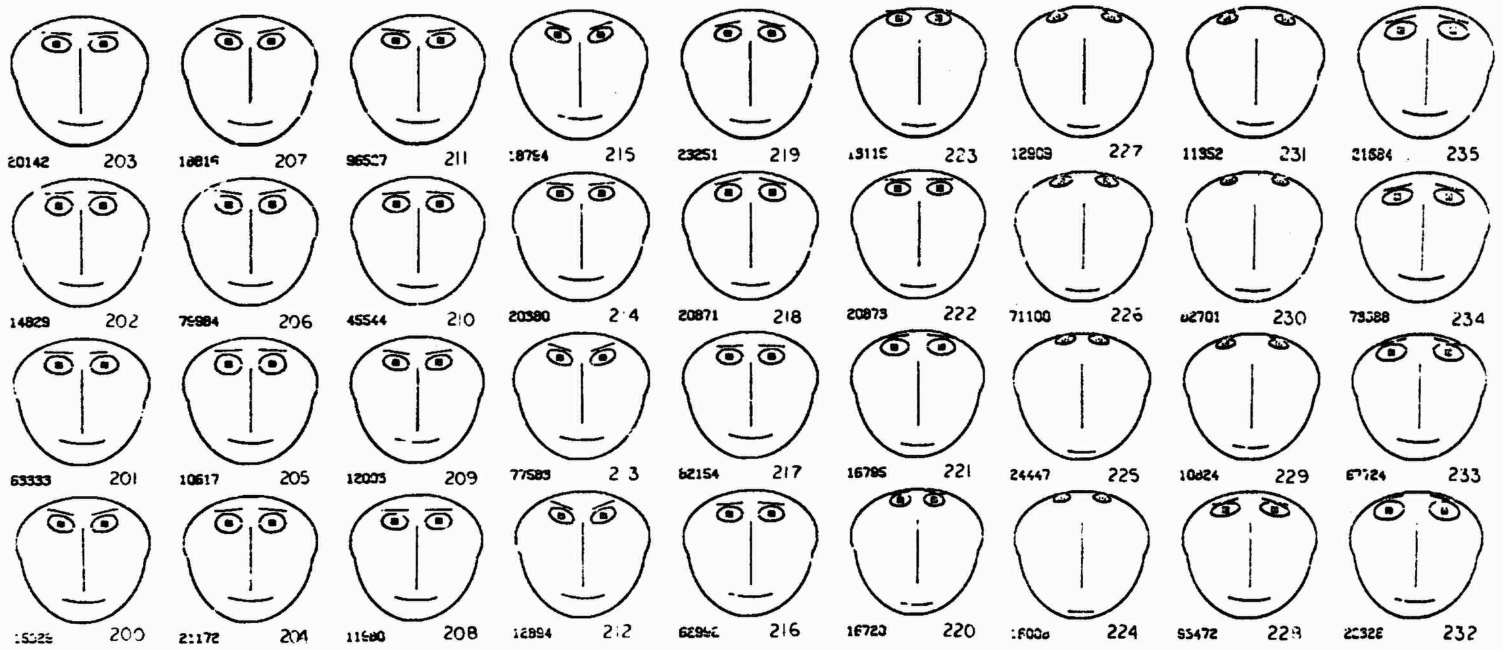
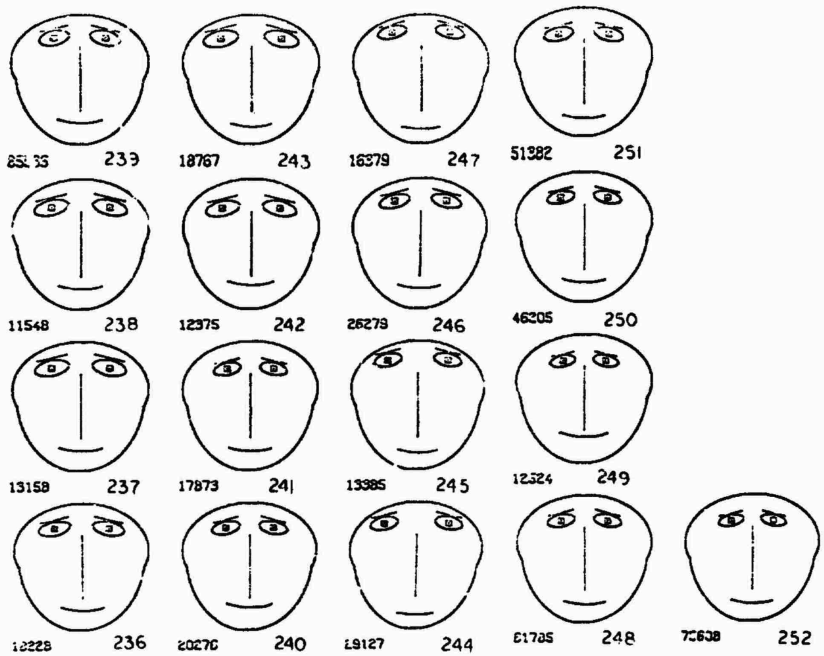

20142 203
18816 207
96527 211
18754 215
23251 219
19115 223
12909 227
11352 231
21684 235
14829 202
75984 206
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20380 214
20871 218
20873 222
71100 226
12701 230
73388 234
63333 201
10617 205
12005 209
77583 213
62154 217
16785 221
24447 225
10824 229
67124 233
15525 200
21172 204
11580 208
12884 212
62952 216
16720 220
16006 224
65472 228
22328 232

FIG. 2F


85133 239
18767 243
16379 247
51382 251
11548 238
12375 242
26279 246
46205 250
13158 237
17873 241
13385 245
12324 249
12228 236
20270 240
19127 244
61785 248
70638 252

contents by one method and repeated assays of 5 of these by a second method. These 12 variables were observed on each of 53 equally spaced specimens along the core and are presented in Table 2a of Appendix A4.

The 53 faces obtained are shown in Figure 2a in the sequence as marked. They clearly indicate the sequence number where certain critical changes take place. One substantial change begins to take place after the 20th specimen, and those from 25 to 32 are quite distinct from the others. Another substantial change evolves from speciments 32 to 35. Particularly characteristic of the group from 25 to 32 are the tiny and high eyes, round face, broad smile, with mouth close to the relatively long nose. The group from 36 to 53 are characterized by a different constellation of special features, suggesting that a traditional linear analysis of this 12-dimensional time series may disguise some of the phenomena clearly observable.

In a casually designed experiment to determine (1) whether the large number of variables used interfered with comprehension, and (2) whether certain features had more impact than others, two additional sequences of faces were run. In Figure 2b, only the first seven variables were used. In Figure 2c, only the last 5 variables were used. These variables, which mainly controlled the eyes in Figure 2a, were made to control the face and mouth. A mistake gave the mouth too large a range, resulting in some peculiar idiosyncracies. The results seemed to indicate that the additional variables add richness to the picture and seem to help the viewer. This cannot be regarded as a serious test, especially since the last 5 variables are supposed to measure 5 of the quantities in the first 7.

Mr. Elliott, the geologist who provided me with the data, seemed to feel that the shape of face carried the essential information and that there was an element of luck in the particular choice of facial parameters selected to be controlled by the variables. He was challenged to make selections which he felt would be least informative. This resulted in Figures 2d, 2e and 2f for the 12, first 7 and last 5 variables respectively.

3. Potential Advantages

Graphical representations have many uses. These include (1) enhancing the user's ability to detect and comprehend important phenomena, (2) serving as a mnemonic device for remembering major conclusions, (3) communicating major conclusions to others, and (4) providing the facility of doing relatively accurate calculations informally. The representation by faces seems to have potential in the first two of these uses.

People grow up studying and reacting to faces all of the time. Small and barely measurable differences are easily detected and evoke emotional reactions from a long catalogue buried in the memory. Relatively large differences go unnoticed in circumstances where they are not important. This implies that the human mind subconsciously operates as a high-speed computer, filtering out insignificant visual phenomena and focusing on the potentially important. Particularly valuable is this flexibility in disregarding non-informative data and searching for useful information. It is this flexibility which is lacking in canned computer programs.

Moreover, this ability is great when applied to the study of faces. Experience with caricatures and cartoons would seem to indicate that

the need for realistic faces on pictures is not great and that lack of realism is compensated, at least in part, by the ability to caricaturize.

The ability to relate faces to emotional reactions seems to carry a mnemonic advantage. For example, in looking at the numerical data from the geological problem, major changes in individual variables are readily apparent. The author found that when studying these numerical data visually with no background in the scientific problem, many changes were observed but attention would be distracted quickly by other effects. After a substantial time, a confusion of reactions remained with little useful memory. Certain major characteristics of the faces are instantly observed and easily remembered in terms of emotions and appearance. Finer details and correlations become apparent after studying the faces for a time. The awareness of these does not drive out of mind the original major impressions.

I would anticipate that the faces would have relatively little usefulness as a communication device. If results from a study of data were translated from the data to the faces, then the mnemonic advantages of the faces could conceivably make it desirable to use faces to communicate a relatively large assortment of results of varying degrees of importance.

Anyone who uses graph paper to analyze data is aware of how precise one can be with rough drawings which are strategically arranged. It would seem that the faces should not be expected to be useful except in the grossest types of calculation.

## 4. Alternative Representations

One is led to ask two questions. First, if this simple idea is so good, why wasn't it thought of before? Second, what alternative representations are there for points in high-dimensional space?

Introspection would suggest that this idea must have been considered before in a simpler form. However, the effective application in this form would require a computer technology which has only recently become available. Thus it is unlikely to have been used in this or similar form in spite of the growing need for a useful representation.

Several more primitive versions have come to my attention. Anderson [1] developed a method of using "glyphs", which are circles of fixed radius with rays of various lengths and directions extending from the boundary. The length of the ray represents the value of a variable. Pickett and White [4] used triangles which represent 4 variables (the three lengths of the sides and the orientation[1]). Both the glyphs and triangles can raise the dimensionality by 2 by locating the center on a point in two-dimensional space. I have some memory of being told of a scheme to convert cardiograms or brain waves to sound in the hope that the human processing of sound would be more revealing than looking at graphs. This idea seems interesting, but no follow-up has come to my attention.

Several alternative representations have been considered. The most standard is the use of profiles. Here one represents a point in k-dimen-

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[1] I have felt greatly indebted to Pickett for many conversations we had in which he emphasized how the human ability to process subconsciously large amounts of information of textures is fundamental to the ability to locomote and, indeed, to exist. After having developed the faces, I noticed his paper containing the triangle representation and realized that I had seen it before but had not paid special attention to it in the form presented.

sional space by a series of $k$ bars at heights corresponding to the values of the variables. It would seem desirable to standardize each variable so that the ranges either go from 0 to 1 or center about the mean. In some variations the bars are replaced by a polygonal line.

A relatively novel variation of the profile method is one due to Daetz [3] where a circle is drawn and along $k$ equally spaced rays from the center, points are marked whose distance from the circumference are equal to standardized distances from the means of the $k$ variables. These points are connected to form a polygon.

The polygons resulting from this variation seem to be more readily translatable to human experience than the simpler profiles. They assume "meaningful" shapes, and the tendency to lean in certain directions has mnemonic force.

A new technique of Andrews [2] consists of generating a Fourier Series of the form

$$f(t) = \frac{x_1}{\sqrt{2}} + x_2 \cos t + x_3 \sin t + x_4 \cos 2t + \cdots$$

where the $x_i$ are the observed variables. This method has the interesting property that if $x$ generates $f$, and $y$ generates $g$, then

$$\int_0^1 [f(t)-g(t)]^2 dt = \sum_{i=1}^{k} (x_i - y_i)^2 \quad ,$$

suggesting that the method could be useful for expressing moderately refined calculations relevant to linear analysis. Andrews has applied this method using the principal components, in place of the original

observations, for the $x_i$ . In the normal multivariate model, these $x_i$ would be independent, and the distances in the above expression would be quite meaningful. A value of t which consistently and widely separates the f(t) of two classes of observations provides an effective linear function for discriminating between the two classes.

It seems reasonable to conjecture that one may, in the spirit of Daetz, achieve more suggestive curves by plotting (f(t) + C,t) in polar coordinates.

## 5. Summary

The use of the face representation provides a promising approach for a first look at multivariate data which is effective in revealing rather complex relations not always visible from simple correlations based on two-dimensional linear theories. It can be used to aid in cluster analysis, discrimination analysis, and to detect substantial changes in time series.

The study of faces does not seem to become more difficult as the number of variables increases. Example 2 indicated that the information content transmitted seems to become richer as the number of variables increases. At this point, one can treat up to 18 variables[1], but it would be relatively easy to increase that number by adding other features such as ears, hair, facial lines, and even possibly by taking pairs of faces.

This approach is an amusing reversal of a common one in artificial intelligence. Instead of using machines to discriminate between human

[1] We shall note in the appendix that the normalization of the width and length of the faces almost eliminates two of these variables.

faces by reducing them to numbers, we discriminate between numbers by using the machine to do the brute labor of drawing faces and leaving the intelligence to the humans, who are still more flexible and clever.

One question frequently asked is whether some features are more informative than others. The individuals who worked on Example 1 felt that one only looked at eyes. Elliott was convinced that only the shapes of the head are relevant. In my opinion, the human will tend to concentrate on what is important in the data. However, this question requires serious study. At present an experiment is under way to determine whether permuting the variables has an effect on the ability of subjects to separate data from a mixture of the two normal multivariate distributions into the appropriate families.

In the meantime, there are a few obvious limitations which require care. When the eyes are very small, the position of the pupil becomes hard to detect. The zero point in the variable which controls the curvature of the mouth may have unusual significance and hence has been avoided in some studies. The corner points where the ellipses of the face meet disappears when the face is circular, losing some information. These are minor points and can easily be avoided.

While the method looks promising, it still remains to be seen whether it can produce results not easily obtained by standard computations on the part of an investigator well versed in statistics and the field of application. One minor success was on the clustering of a randomly selected subset of Fisher's iris data which yielded poor results under the King stepwise clustering algorithm [5]. However, nothing that I would regard as a convincing major success for this method has yet been obtained.

## Appendix

### A1. Construction of Faces

Given 18 numbers $(x_1, x_2, \ldots, x_{18})$ in appropriate ranges (which will usually be 0 to 1), we define a face (see Fig. 3) as follows. Let H be a nominal distance and let $h^* = \frac{1}{2}(1+x_1)H$ be the distance from the origin to a "corner" point P. As $x_1$ varies from 0 to 1, $h^*$ varies from H/2 to H. Let $\theta^* = (2x_2 - 1)\pi/4$ be the angle of OP with the horizontal. Let P' be a point symmetric to P about the vertical axis through 0. Let $h = \frac{1}{2}(1+x_3)H$ represent the distance from 0 to U the top of the head and L the bottom of the head, both on the vertical line through 0. The upper part of the head is an ellipse which is determined by P', U, and P and an eccentricity $x_4$. Let $x_4$ represent the ratio of the width to height of the upper ellipse. Similarly, $x_5$ is the same ratio for the ellipse through P', L, and P. The nose is a vertical line of length $2hx_6$ with 0 as center. The mouth intersects the vertical line extended through the nose at a point $P_m$ whose distance below 0 is $h[x_7 + (1-x_7)x_6]$. This represents a point $x_7$ part of the way from the bottom of the nose to U. The mouth is part of a circle whose center is $h/x_8$ above $P_m$. Thus a positive value of $x_8$ yields a smile. The mouth is symmetric about the vertical axis through 0. Its projection on the horizontal axis has the half-length $a_m = x_9(h/|x_8|)$ unless $(h/|x_8|)$ exceeds the half-width $w_m$ of the face at the height of $P_m$. In that case $x_9 w_m$ is used. The eyes are located at a height $y_e = h[x_{10} + (1-x_{10})x_6]$ above 0 and at centers which are $x_e = w_e(1+2x_{11})/4$ from the vertical axis

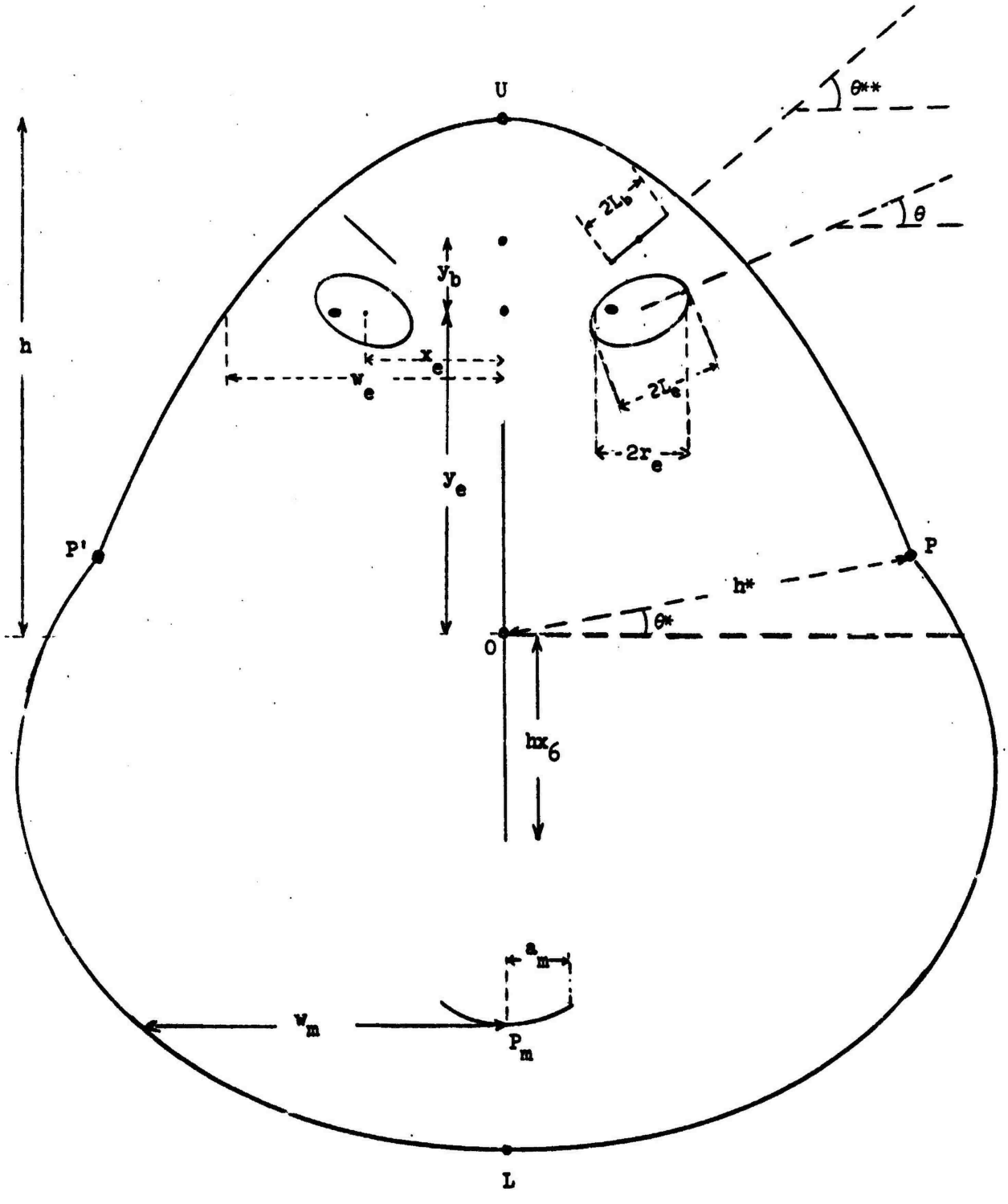



Figure 3

where $w_e$ is the half-width of the face at the height $y_e$. They are symmetrically slanted at an angle $\theta = (2x_{12}-1)\pi/5$ with the horizontal. The eyes are ellipses with eccentricity $x_{13}$ (height/length before slanting) and half-length $L_e = x_{14}\min(x_e, w_e - x_e)$.

The only asymmetry appears in the location of the pupils which move together an amount $r_e(2x_{15}-1)$ from the center of the eye where $r_e = (\cos^2\theta + \sin^2\theta/x_{13}^2)^{-1/2}L_e$ is the horizontal half-length of the slanted eye at height $y_e$.

Finally the eyebrows are symmetrically located with centers at a height $y_b = 2(x_{16}+.3)L_e x_{13}$ above the eye centers and slant $2(x_{17}-1)\pi/5$ with respect to the eye, i.e., $\theta^{**} = \theta + (2x_{17}-1)\pi/5$ with respect to the horizontal and half-length $L_b = r_e(2x_{18}+1)/2$.

One final step taken by the programmer and which has been left intact, is to normalize both horizontal and vertical axes, each by a multiplicative factor, so that the width of the head at its widest part and its height are both equal to a specified constant. This step, which essentially removes two degrees of freedom, was left unaltered for intuitive and aesthetic reasons that are somewhat vague and may require reconsideration when dealing with 18-dimensional data. In the meantime, the effects of $x_1$ and $x_3$ are almost but not completely eliminated because of the secondary effects of the normalization, which will adjust all of the other features at the same time as the width and height are normalized.

Most of the parameters $x_i$ are adjusted to range within a subinterval of $(0,1)$. The exceptions are two of the eccentricities, $x_4$ and $x_5$, and the parameter controlling curvature of the mouth, $x_8$.

Ordinarily $x_4$ and $x_5$ are kept within 1/2 to 2, and $x_8$ is kept within $(-5,5)$ . The eccentricity of the eye $x_{13}$ has usually been kept within $(.4,.8)$ . Some of the ranges must be controlled carefully. We do not want negative length eyes. Others need not be so carefully controlled. It is no calamity to have eyes extend beyond the face.

When the two ellipses of the head meet smoothly, the corner point P is lost, and the variable $x_2$ loses effect. Restricting $x_4$ and $x_5$ to widely separated ranges seems to avoid this problem.

Data are converted to the x parameters as follows. If the variable Z is used to control the parameter $x_i$, which is to be allowed to range from $a_i$ to $b_i$, we let

$$x_i = a_i + (b_i - a_i)\left(\frac{Z - m}{M - m}\right)$$

where m and M are the observed minimum and maximum of Z .

## A2. Formulae Used on the Construction

We describe a few of the less trivial formulae used in the construction of the faces.

The point $P$ has coordinates $x_o = h^* \cos \theta^*$ and $y_o = h^* \sin \theta^*$. The ellipse through $PUP'$ has equation

$$\frac{x^2}{a_u^2} + \frac{(y - c_u)^2}{b_u^2} = 1$$

where $b_u = h - c_u$, $a_u = x_4 b_u$ and

$$c_u = \frac{1}{2}\left[(h+y_o) - \frac{x_o^2}{x_4^2(h - y_o)}\right] .$$

The ellipse through $PLP'$ has equation

$$\frac{x^2}{a_L^2} + \frac{(y-c_L)^2}{b_L^2} = 1$$

where $b_L = h + c_L$, $a_L = x_5 b_L$ and

$$c_L = \frac{1}{2}\left[(-h+y_o) - \frac{x_o^2}{x_5^2(-h-y_o)}\right]$$

The head is then described by $(\pm x(y), y)$ where

$$x(y) = x_4[b_u^2 - (y-c_u)^2]^{1/2} \qquad y_o \le y \le h$$

$$= x_5[b_L^2 - (y-c_L)^2]^{1/2} \qquad -h \le y \le y_o$$

The mouth is a circular arc with curvature $|x_8/h|$ through $(0,y_m)$ where $y_m = -h(x_7 + (1-x_7)x_6)$ . It is described by

$$y = y_m + (\operatorname{sgn} x_8)\left[\frac{h}{|x_8|} - \sqrt{\left(\frac{h}{x_8}\right)^2 - x^2}\right] , \qquad 0 \le x \le a_m$$

where

$$a_m = x_9 \min[x(y_m), h/|x_8|] .$$

The eyes are nominally centered at $(x_e,y_e)$ where

$$y_e = h[x_{10} + (1-x_{10})x_6]$$

$$x_e = x(y_e)[1 + 2x_{11}]/4$$

and have half-length

$$L_e = x_{14} \min[x_e, x(y_e) - x_e]$$

Let $(u,v)$ be the coordinates of an ellipse with center at the origin, half-length $L_e$ and eccentricity $x_{13}$ . Then $v = x_{13}(L^2 - u^2)^{1/2}$ describes part of the ellipse. A similar part of the slanted eye can be described for $0 \le u \le L$ by

$$x = x_e + u \cos \theta - v \sin \theta$$

$$y = y_e + u \sin \theta + v \cos \theta$$

and symmetry is used to complete both eyes.

To place the pupils within the eyes, both are moved a distance $r_e(2x_{15} - 1)$ from the center of the eye, where $r_e$, the horizontal half-length of the slanted eye at height $y_e$, is $(u^2 + v^2)^{1/2}$ when $v/u = \tan \theta$. This yields

$$r_e = L_e(\cos^2 \theta + x_{13}^{-2} \sin^2 \theta)^{-1/2}$$

The program then normalizes all heights and widths by multiplicative factor $k/h$ and $k/\max x(y)$ respectively. Currently $k$ is set at 2 inches. A copy of the program follows.

```
//FACES JOB (J683,323,1.0,41,'BETTY'
//STEP1 EXEC ASMGC
//ASM.SYSIN DD *
RANDOM   CSECT
         USING *,15
         STM   2,5,28(13)
         LM    2,3,0(1)
         L     5,A
         M     4,0(2)
         D     4,P
         ST    4,0(2)
         SRL   4,7
         A     4,CHAR
         ST    4,0(3)
         LM    2,5,28(13)
         BR    14
CHAR     DC    F'1073741824'
A        DC    F'16807'
P        DC    F'2147483647'
         END
/*
//STEP2 EXEC FORTHCL
//FORT.SYSIN DD *
C      PROGRAM DRFACE ON SYS09
       REAL*4 XFACE(1000),YFACE(1000),CAPH,SMALLH,XO,YO,HSTAR,THSTAR,CU,
      *BU,AU,CL,BL,AL,LHSRHS(400),YSAME(201),XPLUS,BUSQ,BLSQ,AN,XNOSE(51)
      *,YNOSE(51),XMOUTH(51),YMOUTH(51),AM,YM,AXB,XLEYE(80),XREYE(80),XE,
      *YE,THETA,X13,L,LSQ,V,XSTAR,YSTAR,YEYES(80),SINTH,COSTH,PUPILX(2),
      *PUPILY(2),R,YB,THSTST,LB,XX,YY,XRBROW(41),XLBROW(41),YRBROWS(41),
      *DATA(18),AMODES(200),ID1(100),
     *Y(19),MINY(19),RANGEY(18),MAXY(18),AI(18),BI(18),BIMAI(18)
       INTEGER*4 IFIX(18),IRAND(18)
       LOGICAL*4 IFMT(18)
       REAL*4 PI/3.141593/
       DATA CAPH/9.0/
       INTEGER*4 NU,NL,NFACE,NNOSE,NMOUTH,NEYES,NBROWS
      *,ISTR,IDENT(100)
       DATA NU,NL,NNOSE,NMOUTH,NEYES,NBROWS/400,400,51,51,80,41/
C
       WRITE(6,402)
  402  FORMAT('1  ID(1)     INTEGER')
       CALL MODESG(AMODES,'BETTY, BIN 323',14)
       READ(5,405)IFMT
       READ(5,1)NPLOTS
     1 FORMAT(1814)
       NINCHS = NPLOTS/4 + 3
       AMODES(97)=250.0*NINCHS
C
C      READ IN STARTING PT. FOR RAND. NO. GENERATOR, IDENT. INFO
       READ(5,405)IFMT
       READ(5,406)INTRAN,IFID,IDNO
  406  FORMAT(I10,2I5)
C
C      READ IN ORDER NUMBERS OF VARIABLES THAT ARE FIXED
       READ(5,405)IFMT
       READ(5,1)NFIXED
       IF(NFIXED.EQ.0)GO TO 1122
       READ(5,1)(IFIX(I),I=1,NFIXED)
       READ(5,2)(DATA(IFIX(I)),I=1,NFIXED)
C
C      READ IN ORDER NUMBERS OF VARIABLES THAT ARE RANDOM
 1122  READ(5,405)IFMT
       READ(5,1)NRAND
       READ(5,1)(IRAND(I),I=1,NRAND)
C
C      CALCULATE NUMBER OF Y VALUES TO READ
       IF(IFID.EQ.0)GO TO 407
       NVREAD=NRAND+1
       GO TO 408
  407  NVREAD=NRAND
       IDCNT=IDNO-1
C
C      READ IN RANGE FOR RANDOM VALUES
  408  READ(5,405)IFMT
       DO 403 I=1,NRAND
       READ(5,2)AI(I),BI(I)
  403  BIMAI(I)=BI(I)-AI(I)
C
C      READ IN MINIMUM AND MAXIMUM OF RANDOM VALUES
       READ(5,405)IFMT
       DO 400 I=1,NRAND
       READ(5,2)MINY(I),MAXY(I)
       RANGEY(I)=MAXY(I)-MINY(I)
  400  CONTINUE
C
C      READ IN FORMAT FOR DATA
       READ(5,405)IFMT
       READ(5,405)IFMT
  405  FORMAT(18A4)
C
       IPLOT=0
       XRMAX=0.0
       DO 50 JPLOT4=1,NPLOTS,4
       XRMIN=XRMAX
       XRMAX=XRMAX+250.
       YRMIN=0.0
       YRMAX=250.
       JEND=MINO(NPLOTS,JPLOT4+3)
       DO 49 JPLOT1=JPLOT4,JEND
       READ(5,IFMT)(Y(I),I=1,NVREAD)
     2 FORMAT(9F8.2)
C
C      FORM 5-DIGIT NUMBER
       CALL RANDOM(INTRAN,UNIFOR)
       RSTR=INTRAN
       IF(IFID.EQ.0)GO TO 23
       XID1=Y(IDNO)
       IF(IDNO.EQ.NVREAD)GO TO 3
       DO 233 J=IDNO,NRAND
  233  Y(J)=Y(J+1)
       GO TO 3
   23  IDCNT=IDCNT+1
       XID1=IDCNT
C
     3 IF(RSTR.GT.99999.0)GO TO 4
       RSTR=RSTR*10.0
       GO TO 3
     4 RSTR=RSTR/10.0
       IF(RSTR.GT.99999.0)GO TO 4
       ISTR=RSTR
       IF(IPLOT.EQ.0)GO TO 46
    44 DO 45 J=1,IPLOT
       IF(ISTR.NE.IDENT(J))GO TO 445
       ISTR=ISTR+1
       GO TO 44
   445 IF(ISTR.LT.IDENT(J))GO TO 456
    45 CONTINUE
    46 IPLOT=IPLOT+1
```

```
      IDENT(IPLOT)=ISTR
      ID1(IPLOT)=XID1
      GO TO 459
  456 INDX=IPLOT+1
      DO 458 I=J,IPLOT
      INDXM1=INDX-1
      IDENT(INDX)=IDENT(INDXM1)
      ID1(INDX)=ID1(INDXM1)
      INDX=INDXM1
  458 CONTINUE
      IPLOT=IPLOT+1
      IDENT(J)=ISTR
      ID1(J)=XID1
C
C     FORM MISSING X VALUES FROM Y
  459 DO 460 J=1,NRAND
      XJ=(Y(J)-MINY(J))/RANGEY(J)
      DATA(IRAND(J))=XJ*SLMAI(J)+AI(J)
  460 CONTINUE
C
      WRITE(6,401)XID1,INTRAN
C
      HSTAR=.5*(1.0+DATA(1))*CAPH
      THSTAR=(2.0*DATA(2)-1.0)*PI*0.25
      SMALLH=.5*(1.0+DATA(3))*CAPH
      XO=HSTAR*COS(THSTAR)
      YO=HSTAR*SIN(THSTAR)
C
C     DRAW FACE
      CU=.5*(SMALLH+YO-XO**2/(DATA(4)**2*(SMALLH-YO)))
      BU=SMALLH-CU
      AU=DATA(4)*BU
      BUSQ=BU**2
      CL=.5*(-SMALLH+YO-XO**2/(DATA(5)**2*(-SMALLH-YO)))
      BL=SMALLH+CL
      AL=DATA(5)*BL
      BLSQ=BL**2
      XMAX=XO
      NFACE=NU+NL
C
      NUP1=NU+1
      YSAME(1)=YO
      LHSRHS(1)=-XO
      NSTEP=NU/2
      NSTPP1=NSTEP+1
      YSAME(NSTPP1)=SMALLH
      LHSRHS(NSTPP1)=0.0
      STPSIZ=(SMALLH-YO)/NSTEP
      ISTOP=NSTEP-1
C
      DO 5 I=1,ISTOP
      IP1=I+1
      YSAME(IP1)=YO+I*STPSIZ
      NUM1=NUP1-I
      XPLUS=DATA(4)*SQRT(BUSQ-(YSAME(IP1)-CU)**2)
      IF(XPLUS.GT.XMAX)XMAX=XPLUS
      LHSRHS(IP1)=-XPLUS
      LHSRHS(NUM1)=XPLUS
    5 CONTINUE
C
      XFACE(1)=LHSRHS(1)
      YFACE(1)=YSAME(1)
      NUP2=NU+2
      DO 6 I=2,NSTEP
      XFACE(I)=LHSRHS(I)
      YFACE(I)=YSAME(I)
      IX2=NUP2-I
      XFACE(IX2)=LHSRHS(IX2)
    6 YFACE(IX2)=YSAME(I)
      XFACE(NSTPP1)=LHSRHS(NSTPP1)
      YFACE(NSTPP1)=YSAME(NSTPP1)
C
      YSAME(1)=YO
      LHSRHS(1)=XO
      NLP1=NL+1
      YSAME(NSTPP1)=-SMALLH
      LHSRHS(NSTPP1)=0.0
      STPSIZ=(YO+SMALLH)/NSTEP
      DO 7 I=1,ISTOP
      IP1=I+1
      NLMI=NLP1-I
      YSAME(IP1)=YO-I*STPSIZ
      XPLUS=DATA(5)*SQRT(BLSQ-(YSAME(I)-CL)**2)
      IF(XPLUS.GT.XMAX)XMAX=XPLUS
      LHSRHS(IP1)=XPLUS
      LHSRHS(NLMI)=-XPLUS
    7 CONTINUE
C
      NLP2=NL+2
      XFACE(NUP1)=LHSRHS(1)
      YFACE(NUP1)=YSAME(1)
      DO 8 I=2,NSTEP
      XFACE(NU+I)=LHSRHS(I)
      YFACE(NU+I)=YSAME(I)
      IX2=NLP2-I
      XFACE(NU+IX2)=LHSRHS(IX2)
      YFACE(NU+IX2)=YSAME(I)
    8 CONTINUE
      XFACE(NU+NSTPP1)=LHSRHS(NSTPP1)
      YFACE(NU+NSTPP1)=YSAME(NSTPP1)
      XMIN=-XMAX
      YMAX=SMALLH
      YMIN=-SMALLH
C
C     DRAW NOSE
      AN=SMALLH*DATA(6)
      XNOSE(1)=0.0
      XNOSE(2)=0.0
      YNOSE(1)=AN
      YNOSE(2)=-AN
C
C     DRAW MOUTH
      YM=-SMALLH*(DATA(6)+(1.0-DATA(6))*DATA(7))
      XOFYM=DATA(5)*SQRT(BLSQ-(YM-CL)**2)
      AX8=SMALLH/ABS(DATA(8))
      AM=DATA(9)*AMIN1(XOFYM,AX8)
      NSTEP=NMOUTH/2
      NMP1=NMOUTH+1
      YMOUTH(NSTEP+1)=YM
      XMOUTH(NSTEP+1)=0.0
      STPSIZ=AM/NSTEP
      X8SQ=(SMALLH/DATA(8))**2
      HBY8=AX8
      IF(DATA(8).LT.0.0)SIGN=-1.0
      IF(DATA(8).GT.0.0)SIGN=1.0
      DO 11 I=1,NSTEP
      XPLUS=-AM+(I-1)*STPSIZ
      XMOUTH(I)=XPLUS
      NMMI=NMP1-I
      XMOUTH(NMMI)=-XPLUS
      YMOUTH(I)=YM+SIGN*(HBY8    -SQRT(X8SQ-XPLUS**2))
   11 YMOUTH(NMMI)=YMOUTH(I)
```

```
C
C     DRAW EYES
      YE=SMALLH*(DATA(6)+(1.0-DATA(6))*DATA(10))
      XOFYE=DATA(4)*SQRT(BUSQ-(YE-CU)**2)
      XE=XOFYE*(1.0+2.0*DATA(11))*0.25
      THETA=(2.0*DATA(12)-1.0)*PI*0.2
      X13=DATA(13)
      L=DATA(14)*AMIN1(XE,XOFYE-XE)
      LSQ=L**2
      SINTH=SIN(THETA)
      COSTH=COS(THETA)
      R=L/SQRT(COSTH**2+SINTH**2/X13**2)
      PUPILX(1)=-XE+R*(2.0*DATA(15)-1.0)
      PUPILX(2)=XE+R*(2.0*DATA(15)-1.0)
      PUPILY(1)=YE
      PUPILY(2)=YE
C
      NSTEP=NEYES/4
      STPSIZ=L/NSTEP
      I1=1
      I2=NSTEP+1
      I3=2*NSTEP+1
      I4=3*NSTEP+1
      U=0.0
      V=X13*L
      XSTAR=-V*SINTH
      YSTAR=V*COSTH
      XX=XE+XSTAR
      YY=YE+YSTAR
      XREYE(I2)=XX
      XLEYE(I2)=-XX
      YEYES(I2)=YY
      XX=XE-XSTAR
      YY=YE-YSTAR
      XREYE(I4)=XX
      XLEYE(I4)=-XX
      YEYES(I4)=YY
      U=L
      XSTAR=U*COSTH
      YSTAR=U*SINTH
      XX=XE+XSTAR
      YY=YE+YSTAR
      XREYE(I3)=XX
      XLEYE(I3)=-XX
      YEYES(I3)=YY
      XX=XE-XSTAR
      YY=YE-YSTAR
      XREYE(I1)=XX
      XLEYE(I1)=-XX
      YEYES(I1)=YY
      I1=I2
      I2=I4
      ISTOP=NSTEP-1
      DO 12 I=1,ISTOP
      U=I*STPSIZ
      V=X13*SQRT(LSQ-U**2)
      XSTAR=U*COSTH-V*SINTH
      YSTAR=U*SINTH+V*COSTH
      XX=XE+XSTAR
      YY=YE+YSTAR
      I2=I2+1
      I4=I4+1
      XREYE(I2)=XX
      XLEYE(I2)=-XX
      YEYES(I2)=YY
      XX=XE-XSTAR
      YY=YE-YSTAR
      XREYE(I4)=XX
      XLEYE(I4)=-XX
      YEYES(I4)=YY
      XSTAR=U*COSTH+V*SINTH
      YSTAR=U*SINTH-V*COSTH
      I1=I1-1
      I3=I3-1
      XX=XE-XSTAR
      YY=YE-YSTAR
      XREYE(I1)=XX
      XLEYE(I1)=-XX
      YEYES(I1)=YY
      XX=XE+XSTAR
      YY=YE+YSTAR
      XREYE(I3)=XX
      XLEYE(I3)=-XX
      YEYES(I3)=YY
   12 CONTINUE
C
C     DRAW EYEBROWS
      YB=YE+2.0*(0.3 + DATA(16))  *L*X13
      THSTST=THETA+PI*(2.0*DATA(17)-1.0)*0.2
      COSTH=COS(THSTST)
      SINTH=SIN(THSTST)
      LB=R*(2.0*DATA(18)+1.0)*0.5
      XX=LB*COSTH+XE
      YY=LB*SINTH+YB
      XRBROW(1)=XX
      XLBROW(1) =-XX
      YBROWS(1) =YY
      XX=-LB*COSTH+XE
      YY=-LB*SINTH+YB
      XRBROW(2) =XX
      XLBROW(2) =-XX
      YBROWS(2) =YY
C
C     ADJUST X AND Y MIN AND MAX TO ALLOW FOR MARGINS
      XLAB=XMIN
      DEL1=(XMAX-XMIN)/8.0
      XMIN=XMIN-DEL1
      XMAX=XMAX+DEL1
      DEL2=(YMAX-YMIN)/8.0
      YLAB=YMIN-DEL2
      YMIN=YMIN-2.0*DEL2
C
C     DRAW CURVES ON CALCOMP
      CALL SUBJEG(AMODES,XMIN,YMIN,XMAX,YMAX)
      CALL OBJECG(AMODES,XMMIN,YMMIN,XMMAX,YMMAX)
C
C     LABEL PLOT WITH 5 DIGIT NUMBER
      CALL INUMPG(AMODES,XLAB,YLAB,5,IST,1)
C
C     DRAW FACE
      CALL LINESG(AMODES,NU,XFACE,YFACE)
      CALL LINESG(AMODES,NL,XFACE(NUP1),YFACE(NUP1))
C
C     NOSE
      CALL LINEG(AMODES,0,XNOSE(1),YNOSE(1))
      CALL LINEG(AMODES,1,XNOSE(2),YNOSE(2))
C
C     MOUTH
      CALL LINESG(AMODES,NMOUTH,XMOUTH,YMOUTH)
C
C     EYES
      CALL LINESG(AMODES,NEYES,XLEYE,YEYES)
```

```
      CALL LINESG(AMODES,NEYES,XREYE,YEYES)
C
C     EYEBROWS
      CALL LINEG(AMODES,0,XRBROW(1),YBROWS(1))
      CALL LINEG(AMODES,1,XRBROW(2),YBROWS(2))
      CALL LINEG(AMODES,0,XLBROW(1),YBROWS(1))
      CALL LINEG(AMODES,1,XLBPOW(2),YBROWS(2))
C
C     PUPILS
      CALL POINTG(AMODES,2,PUPILX,PUPILY)
      YRMIN=YRMAX
      YRMAX=YRMAX+250.
   49 CONTINUE
   50 CONTINUE
C
      WRITE(6,402)
  401 FORMAT(F10.3,2X,I10)
      WRITE(6,401)(ID1(J),IDENT(J),J=1,NPLOTS)
C
      CALL EXITG(AMODES)
C
      STOP
      END
/*
//FT16F001 DD UNIT=2314,VOLUME=SER=SYS03,DISP=(NEW,PASS),          XXX
//            DCB=(RECFM=F,BLKSIZE=600),SPACE=(TRK,(15,5),RLSE)
//PLOTTAPE DD DSNAME=PLOTTAPE,DISP=(NEW,KEEP),VOLUME=PRIVATE,      XXX
//            UNIT=TAPE7,LABEL=(,BLP)
//LKED.SYSLMOD DD DSNAME=J683.LIBRARY(FACES),UNIT=2314,            XXX
//            SPACE=(TRK,(10,5,1),RLSE),VOLUME=SER=SYS04,          XXX
//            DISP=(NEW,KEEP)
/*
```

A3. Dictionary of Parameters and Features They Control

The following table provides a dictionary and ranges within which the $x_i$ are typically restrained.

Table 1

| Range | | | | |
|---|---|---|---|---|
| (0,1) | $x_1$ | constrols | $h^*$ | distance from O to P |
| (0,1) | $x_2$ | constrols | $\theta^*$ | angle between OP and horizontal |
| (0,1) | $x_3$ | controls | h | half-height of face |
| (0.5,2) | $x_4$ | is | | eccentricity of upper ellipse of face (width/height) |
| (0.5,2) | $x_5$ | is | | eccentricity of lower ellipse of face (width/height) |
| (0,1) | $x_6$ | controls | | length of nose |
| (0,1) | $x_7$ | controls | $P_m$ | position of center of mouth |
| (-5,5) | $x_8$ | controls | | curvature of mouth (radius = $h/x_8$) |
| (0,1) | $x_9$ | controls | $a_m$ | length of mouth |
| (0,1) | $x_{10}$ | controls | $y_e$ | height of centers of eyes |
| (0,1) | $x_{11}$ | controls | $x_e$ | separation of centers of eyes |
| (0,1) | $x_{12}$ | controls | $\theta$ | slant of eyes |
| (0.4,0.8) | $x_{13}$ | is | | eccentricity of eyes (height/width) |
| (0,1) | $x_{14}$ | controls | $L_e$ | half-length of eye ($L_e$ also depends in part on $x_{10}$ and $x_{11}$) |
| (0,1) | $x_{15}$ | controls | | position of pupils |
| (0,1) | $x_{16}$ | controls | $y_b$ | height of eyebrow center relative to eye |
| (0,1) | $x_{17}$ | controls | $\theta^{**}-\theta$ | angle of brow relative to eye |
| (0,1) | $x_{18}$ | controls | | length of brow |

A4. Data for Examples

Example 1: In Table 2a we present a list of variables $Z_1, Z_2, \ldots, Z_6$ preceded by a specimen number. At the end of the list are appended the minima $m_i$ and maxima $M_i$ for the six variables $Z_i$, i=1,2,...,6 used in the faces. The 34th specimen was omitted from the faces because an error in copying had made it seem unreliable.

(In a second study, the 40 specimens from 1 to 41, omitting 34, were used. The minima and maxima used for that study are listed as $m_i^*$, $M_i^*$ .)

The variables are measurements of 87 nummulited specimens from the Eocene Yellow Limestone Formation, Jamaica [6]. They represent

$Z_1$ = inner diameter of embryonic chamber (in microns)

$Z_2$ = total number of whorls

$Z_3$ = number of chambers in first whorl

$Z_4$ = number of chambers in last whorl

$Z_5$ = maximum height of chambers in first whorl (in microns)

$Z_6$ = maximum height of chambers in last whorl (in microns)

Table 2b identifies the feature variables controlled by the data and the ranges $(a_i, b_i)$ which correspond to the minima $(m_i, M_i)$ in the first set of faces and $(m_i^*, M_i^*)$ in the second set. Tables 2c and 2d are the dictionary relating specimen identity (i.d.) numbers to random code numbers for the two studies.

Table 2a

6 Measurements on 87 Nummulited Specimens

from the Eocene Yellow Limestone Formation, Jamaica

| ID | $Z_1$ | $Z_2$ | $Z_3$ | $Z_4$ | $Z_5$ | $Z_6$ |
|---|---|---|---|---|---|---|
| 1 | 160 | 51 | 10 | 28 | 70 | 450 |
| 2 | 155 | 52 | 8 | 27 | 85 | 400 |
| 3 | 141 | 49 | 11 | 25 | 72 | 380 |
| 4 | 130 | 50 | 10 | 26 | 75 | 560 |
| 5 | 161 | 50 | 10 | 27 | 70 | 665 |
| 6 | 135 | 50 | 10 | 27 | 88 | 570 |
| 7 | 165 | 50 | 11 | 23 | 95 | 675 |
| 8 | 150 | 50 | 9 | 29 | 90 | 580 |
| 9 | 148 | 48 | 8 | 26 | 85 | 390 |
| 10 | 150 | 45 | 7 | 31 | 60 | 435 |
| 11 | 120 | 40 | 6 | 33 | 55 | 440 |
| 12 | 120 | 51 | 8 | 32 | 56 | 650 |
| 13 | 100 | 42 | 8 | 30 | 55 | 640 |
| 14 | 100 | 44 | 9 | 35 | 48 | 430 |
| 15 | 150 | 40 | 7 | 29 | 65 | 650 |
| 16 | 90 | 45 | 9 | 30 | 70 | 655 |
| 17 | 75 | 42 | 8 | 28 | 60 | 640 |
| 18 | 120 | 47 | 7 | 35 | 67 | 645 |
| 19 | 200 | 43 | 9 | 30 | 62 | 660 |
| 20 | 120 | 41 | 8 | 28 | 63 | 530 |
| 21 | 105 | 50 | 7 | 27 | 64 | 435 |
| 22 | 210 | 52 | 9 | 26 | 67 | 440 |
| 23 | 90 | 40 | 10 | 25 | 68 | 430 |
| 24 | 110 | 52 | 11 | 25 | 60 | 530 |
| 25 | 100 | 43 | 9 | 25 | 70 | 440 |
| 26 | 90 | 44 | 7 | 36 | 63 | 454 |
| 27 | 70 | 45 | 8 | 23 | 64 | 450 |
| 28 | 100 | 48 | 9 | 27 | 65 | 355 |
| 29 | 130 | 52 | 9 | 25 | 70 | 380 |
| 30 | 90 | 45 | 11 | 37 | 74 | 350 |
| 31 | 80 | 46 | 10 | 32 | 78 | 450 |
| 32 | 95 | 49 | 10 | 25 | 82 | 260 |
| 33 | 70 | 44 | 12 | 30 | 85 | 262 |
| 35 | 95 | 51 | 15 | 31 | 70 | 270 |
| 36 | 100 | 46 | 11 | 24 | 76 | 270 |
| 37 | 95 | 48 | 10 | 27 | 74 | 355 |
| 38 | 85 | 47 | 12 | 25 | 73 | 360 |
| 39 | 70 | 48 | 11 | 26 | 78 | 365 |
| 40 | 80 | 54 | 10 | 21 | 80 | 370 |
| 41 | 85 | 55 | 13 | 33 | 81 | 355 |

Table 2a (Cont'd)

| ID | $Z_1$ | $Z_2$ | $Z_3$ | $Z_4$ | $Z_5$ | $Z_6$ |
|---|---|---|---|---|---|---|
| 42 | 200 | 34 | 10 | 24 | 98 | 1210 |
| 43 | 260 | 31 | 8 | 21 | 110 | 1220 |
| 44 | 195 | 30 | 9 | 20 | 105 | 1130 |
| 45 | 195 | 32 | 9 | 19 | 110 | 1010 |
| 46 | 220 | 33 | 10 | 24 | 95 | 1205 |
| 47 | 220 | 30 | 8 | 25 | 90 | 1210 |
| 48 | 190 | 34 | 9 | 26 | 96 | 1070 |
| 49 | 285 | 30 | 11 | 19 | 100 | 990 |
| 50 | 300 | 30 | 9 | 20 | 102 | 1120 |
| 51 | 225 | 30 | 10 | 22 | 105 | 985 |
| 52 | 260 | 34 | 8 | 22 | 97 | 1090 |
| 53 | 280 | 30 | 8 | 20 | 112 | 1200 |
| 54 | 300 | 34 | 10 | 20 | 108 | 835 |
| 55 | 310 | 30 | 11 | 19 | 106 | 1055 |
| 56 | 290 | 31 | 12 | 26 | 94 | 1240 |
| 57 | 260 | 30 | 8 | 22 | 98 | 1015 |
| 58 | 290 | 33 | 9 | 25 | 100 | 1010 |
| 59 | 160 | 31 | 11 | 20 | 79 | 1170 |
| 60 | 240 | 35 | 11 | 20 | 88 | 990 |
| 61 | 195 | 31 | 8 | 21 | 81 | 975 |
| 62 | 290 | 34 | 10 | 19 | 94 | 860 |
| 63 | 210 | 35 | 9 | 22 | 96 | 950 |
| 64 | 180 | 30 | 11 | 22 | 97 | 990 |
| 65 | 205 | 29 | 11 | 23 | 90 | 805 |
| 66 | 215 | 34 | 8 | 21 | 100 | 700 |
| 67 | 270 | 31 | 8 | 20 | 111 | 1170 |
| 68 | 290 | 30 | 9 | 23 | 102 | 1350 |
| 69 | 320 | 32 | 10 | 19 | 87 | 1160 |
| 70 | 210 | 30 | 9 | 18 | 112 | 1010 |
| 71 | 210 | 30 | 8 | 21 | 95 | 1190 |
| 72 | 185 | 34 | 9 | 25 | 96 | 1055 |
| 73 | 200 | 32 | 8 | 26 | 98 | 980 |
| 74 | 170 | 29 | 9 | 20 | 95 | 1095 |
| 75 | 140 | 30 | 9 | 20 | 98 | 990 |
| 76 | 90 | 52 | 8 | 24 | 120 | 210 |
| 77 | 110 | 49 | 9 | 22 | 130 | 220 |
| 78 | 100 | 56 | 8 | 19 | 128 | 216 |
| 79 | 95 | 49 | 8 | 24 | 124 | 218 |
| 80 | 65 | 62 | 9 | 30 | 134 | 200 |
| 81 | 55 | 50 | 10 | 27 | 128 | 205 |
| 82 | 70 | 53 | 7 | 28 | 118 | 204 |
| 83 | 85 | 49 | 11 | 19 | 117 | 206 |
| 84 | 115 | 50 | 10 | 21 | 122 | 198 |
| 85 | 110 | 57 | 9 | 26 | 125 | 230 |
| 86 | 95 | 48 | 8 | 27 | 114 | 228 |
| 87 | 95 | 49 | 8 | 29 | 118 | 240 |
| 88 | 120 | 61 | 9 | 24 | 120 | 244 |
| m | 70 | 29 | 6 | 18 | 48 | 198 |
| M | 320 | 62 | 15 | 37 | 134 | 1350 |
| m* | 70 | 40 | 6 | 21 | 48 | 260 |
| M* | 210 | 55 | 15 | 37 | 95 | 675 |

Table 2b

Details on Faces for Fossil Data

| Feature | First Set of Faces | | Second Set of Faces | |
|---|---|---|---|---|
| | Range | Data | Range | Data |
| $x_1$ | (0.3,0.8) | $Z_1$ | 0.9 | |
| $x_2$ | (0.1,0.5) | $Z_2$ | (0.2,0.8) | $Z_1$ |
| $x_3$ | 0.7 | | 0.9 | |
| $x_4$ | (1.2,2.0) | $Z_3$ | 0.75 | |
| $x_5$ | 1.0 | | (1.0,2.0) | $Z_4$ |
| $x_6$ | 0.3 | | 0.4 | |
| $x_7$ | (0.2,0.8) | $Z_4$ | 0.5 | |
| $x_8$ | (0.5,5.0) | $Z_5$ | (-4.0,4.0) | $Z_3$ |
| $x_9$ | 0.5 | | (0.2,0.8) | $Z_4$ |
| $x_{10}$ | 0.5 | | 0.5 | |
| $x_{11}$ | 0.5 | | 0.5 | |
| $x_{12}$ | 0.5 | | (0.2,0.8) | $Z_5$ |
| $x_{13}$ | 0.6 | | 0.6 | |
| $x_{14}$ | (0.2,0.8) | $Z_6$ | (0.2,0.8) | $Z_6$ |
| $x_{15}$ | 0.5 | | 0.5 | |
| $x_{16}$ | 0.5 | | 0.5 | |
| $x_{17}$ | 0.5 | | 0.5 | |
| $x_{18}$ | 0.5 | | 0.5 | |

Table 2c

Dictionary for 87 Faces in Figure 1A

| ID(1) | Integer | ID(1) | Integer | ID(1) | Integer | ID(1) | Integer |
|---|---|---|---|---|---|---|---|
| 1.000 | 2069660917 | 46.000 | 2069660917 | 43.000 | 10975 | 87.000 | 10875× |
| 2.000 | 1998401560 | 47.000 | 1998401560 | 8.000 | 11475 | 53.000 | 11475× |
| 3.000 | 490779840 | 48.000 | 490779840 | 35.000 | 11518 | 79.000 | 11518× |
| 4.000 | 52082753 | 49.000 | 52082753 | 39.000 | 11675 | 83.000 | 11675× |
| 5.000 | 1328985342 | 50.000 | 1328985342 | 26.000 | 11704 | 71.000 | 11704× |
| 6.000 | 279230547 | 51.000 | 279230547 | 21.000 | 12820× | 66.000 | 12820 |
| 7.000 | 7760347 34 | 52.000 | 7760347 34 | 16.000 | 13111 | 61.000 | 13111× |
| 8.000 | 1147586107 | 53.000 | 1147586107 | 5.000 | 13289× | 50.000 | 13289 |
| 9.000 | 929066642 | 54.000 | 929066642 | 23.000 | 13627× | 68.000 | 13627 |
| 10.000 | 469454757 | 55.000 | 469454757 | 31.000 | 14817× | 76.000 | 14817 |
| 11.000 | 271181821 | 56.000 | 271181821 | 18.000 | 14888 | 63.000 | 14888× |
| 12.000 | 792566613 | 57.000 | 792566613 | 22.000 | 15697 | 67.000 | 15697× |
| 13.000 | 1973485997 | 58.000 | 1973485997 | 24.000 | 16034 | 69.000 | 16034× |
| 14.000 | 494223664 | 59.000 | 494223664 | 45.000 | 16079 | 78.000 | 16128 |
| 15.000 | 2097857899 | 60.000 | 2097857899 | 33.000 | 16128× | 77.000 | 17081× |
| 16.000 | 1311192047 | 61.000 | 1311192047 | 32.000 | 17081 | 62.000 | 18750 |
| 17.000 | 1875032062 | 62.000 | 1875032062 | 17.000 | 18750× | 70.000 | 19083 |
| 18.000 | 1488829956 | 63.000 | 1488829956 | 25.000 | 19083× | 58.000 | 19734 |
| 19.000 | 285615648 | 64.000 | 285615648 | 13.000 | 19734× | 47.000 | 19984× |
| 20.000 | 716244891 | 65.000 | 716244891 | 2.000 | 19984 | 46.000 | 20696 |
| 21.000 | 1282041602 | 66.000 | 1282041602 | 1.000 | 20696× | 80.000 | 20726 |
| 22.000 | 1569774463 | 67.000 | 1569774463 | 36.000 | 20726× | 60.000 | 20978 |
| 23.000 | 1362796246 | 68.000 | 1362796246 | 15.000 | 20978× | 84.000 | 21101 |
| 24.000 | 1603411267 | 69.000 | 1603411267 | 40.000 | 21101× | 85.000 | 26989× |
| 25.000 | 1908361913 | 70.000 | 1908361913 | 41.000 | 26989 | 56.000 | 27118 |
| 26.000 | 1170403846 | 71.000 | 1170403846 | 11.000 | 27118× | 72.000 | 27233 |
| 27.000 | 27233202 | 72.000 | 27233202 | 27.000 | 27233× | 51.000 | 27923× |
| 28.000 | 294409203 | 73.000 | 294409203 | 6.000 | 27923 | 64.000 | 28561 |
| 29.000 | 333152133 | 74.000 | 333152133 | 19.000 | 28561× | 73.000 | 29440× |
| 30.000 | 798031602 | 75.000 | 798031602 | 28.000 | 29440 | 74.000 | 33315 |
| 31.000 | 1481759299 | 76.000 | 1481759299 | 29.000 | 33315× | 55.000 | 46945× |
| 32.000 | 1708167681 | 77.000 | 1708167681 | 10.000 | 46945 | 48.000 | 49077 |
| 33.000 | 1612821471 | 78.000 | 1612821471 | 3.000 | 49077× | 59.000 | 49422× |
| 35.000 | 1151870663 | 79.000 | 1151870663 | 14.000 | 49422 | 81.000 | 51114× |
| 36.000 | 2072638983 | 80.000 | 2072638983 | 37.000 | 51114 | 49.000 | 52082× |
| 37.000 | 511149294 | 81.000 | 511149294 | 4.000 | 52082 | 86.000 | 57887 |
| 38.000 | 951596258 | 82.000 | 951596258 | 42.000 | 57887× | 65.000 | 71624× |
| 39.000 | 1167588997 | 83.000 | 1167588997 | 20.000 | 71624 | 52.000 | 77603 |
| 40.000 | 2110189940 | 84.000 | 2110189940 | 7.000 | 77603× | 88.000 | 78283 |
| 41.000 | 269891375 | 85.000 | 269891375 | 44.000 | 78283× | 57.000 | 79256× |
| 42.000 | 578877161 | 86.000 | 578877161 | 12.000 | 79256 | 75.000 | 79803× |
| 43.000 | 1087524017 | 87.000 | 1087524017 | 30.000 | 79803 | 54.000 | 92906 |
| 44.000 | 782834102 | 88.000 | 782834102 | 9.000 | 92906× | 82.000 | 95159 |
| 45.000 | 1607930792 | | | 38.000 | 95159× | | |

Table 2d

Dictionary for 40 Faces in Figure 1B

| ID(1) | Integer | ID(1) | Integer |
|---|---|---|---|
| 6.000 | 10617 | 1.000 | 1532932388 |
| 30.000 | 10824 | 2.000 | 633332057 |
| 32.000 | 11352 | 3.000 | 1482927467 |
| 40.000 | 11548 | 4.000 | 2014214434 |
| 9.000 | 11980 | 5.000 | 2117264577 |
| 10.000 | 12003 | 6.000 | 1061714849 |
| 13.000 | 12894 | 7.000 | 799844220 |
| 28.000 | 12909 | 8.000 | 1881658967 |
| 24.000 | 13115 | 9.000 | 1198072647 |
| 39.000 | 13158 | 10.000 | 1200303857 |
| 3.000 | 14829 | 11.000 | 45544681 |
| 1.000 | 15329 | 12.000 | 965275235 |
| 25.000 | 16008 | 13.000 | 1289405207 |
| 21.000 | 16720 | 14.000 | 775832172 |
| 22.000 | 16795 | 15.000 | 2038093867 |
| 38.000 | 18228 | 16.000 | 1879453019 |
| 16.000 | 18794 | 17.000 | 629926610 |
| 8.000 | 18816 | 18.000 | 82154560 |
| 4.000 | 20142 | 19.000 | 2087188546 |
| 15.000 | 20380 | 20.000 | 232518877 |
| 19.000 | 20871 | 21.000 | 1672011846 |
| 23.000 | 20873 | 22.000 | 1679574727 |
| 5.000 | 21172 | 23.000 | 2087380521 |
| 37.000 | 21684 | 24.000 | 1311559055 |
| 33.000 | 22326 | 25.000 | 1600884577 |
| 20.000 | 23251 | 26.000 | 244472376 |
| 26.000 | 24447 | 27.000 | 711006721 |
| 11.000 | 45544 | 28.000 | 1290947939 |
| 17.000 | 62992 | 29.000 | 934725132 |
| 2.000 | 63333 | 30.000 | 1082415719 |
| 35.000 | 67724 | 31.000 | 827015496 |
| 27.000 | 71100 | 32.000 | 1135277888 |
| 36.000 | 73688 | 33.000 | 223260021 |
| 14.000 | 77583 | 35.000 | 677241638 |
| 7.000 | 79984 | 36.000 | 736880766 |
| 18.000 | 82154 | 37.000 | 216841913 |
| 31.000 | 82701 | 38.000 | 182282832 |
| 41.000 | 85536 | 39.000 | 1315876802 |
| 29.000 | 93472 | 40.000 | 1154814408 |
| 12.000 | 96527 | 41.000 | 8553670 |

Example 2: Table 3a contains specimen number (i.d. and 12 measurements) for 53 specimens taken at intervals from a 4500-foot core drilled from a Colorado mountainside to locate a deposit of molybdenum. The 12 variables $Z_i$, i=1,2,...,12 represent mineral contents. The last 5 variables are measurements by different methods of 5 of the minerals covered in the first seven measurements. At the bottom of the table are the minima and maxima $m_i$ and $M_i$. Further identification of the data has not been furnished me. When no trace of the element appeared, the nominal value 0.001 was used.

Six different sets of faces were obtained. The feature variables and the ranges $(a_i,b_i)$ corresponding to each set are identified in Table 3b.

Table 3c indicates the two-way dictionary between the specimen number (ID) and the randomly generated code number. In the second part, the first five digits of the code number is presented, but in the first part the code has up to ten digits.

A copy of the program is presented in Table 4. This program was put into a memory file in compiled form and requires a simpler program to enter the relevant data and parameters to drive the main program. This program was constructed by Mrs. Elizabeth Hinkley. At this time the cost of drawing these faces is about 20 to 25 cents per face on the IBM 360-67 at Stanford University using the Calcomp Plotter. Most of that cost is in the computing. I believe that with some attention to cost-cutting it may be possible to reduce this cost considerably. For example, many square roots needed for neighboring points could be replaced

by linear approximations derived from Taylor Expansions based on the preceding point.

I wish to thank Mrs. Hinkley for her excellent work in assembling the program so that it could be used conveniently.

Table 3a

Data on 12 Variables Representing Mineral Contents
From a 4500-Foot Core Drilled from a Colorado Mountainside

| ID | $Z_1$ | $Z_2$ | $Z_3$ | $Z_4$ | $Z_5$ | $Z_6$ | $Z_7$ | $Z_8$ | $Z_9$ | $Z_{10}$ | $Z_{11}$ | $Z_{12}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 200 | 320 | 105 | 057 | 050 | 001 | 001 | 001 | 060 | 020 | 250 | 210 | 370 |
| 201 | 280 | 150 | 040 | 050 | 001 | 001 | 001 | 060 | 040 | 210 | 130 | 420 |
| 202 | 260 | 165 | 033 | 050 | 001 | 001 | 001 | 060 | 010 | 250 | 090 | 440 |
| 203 | 305 | 110 | 044 | 040 | 001 | 001 | 001 | 050 | 050 | 260 | 140 | 250 |
| 204 | 290 | 160 | 035 | 035 | 001 | 001 | 001 | 050 | 020 | 210 | 060 | 510 |
| 205 | 275 | 130 | 047 | 035 | 001 | 001 | 001 | 050 | 020 | 230 | 090 | 570 |
| 206 | 230 | 155 | 035 | 035 | 001 | 001 | 001 | 080 | 020 | 270 | 170 | 400 |
| 207 | 300 | 115 | 050 | 060 | 001 | 001 | 001 | 120 | 010 | 280 | 190 | 300 |
| 208 | 250 | 130 | 041 | 030 | 005 | 001 | 001 | 070 | 030 | 250 | 110 | 330 |
| 209 | 285 | 120 | 047 | 040 | 001 | 001 | 001 | 070 | 010 | 240 | 170 | 280 |
| 210 | 280 | 105 | 047 | 070 | 001 | 001 | 001 | 060 | 020 | 370 | 070 | 300 |
| 211 | 300 | 135 | 050 | 040 | 001 | 001 | 001 | 120 | 060 | 250 | 160 | 200 |
| 212 | 280 | 110 | 056 | 050 | 001 | 001 | 001 | 150 | 010 | 280 | 270 | 280 |
| 213 | 305 | 080 | 065 | 080 | 005 | 001 | 001 | 130 | 010 | 300 | 260 | 260 |
| 214 | 230 | 175 | 029 | 035 | 001 | 001 | 001 | 270 | 030 | 250 | 140 | 240 |
| 215 | 325 | 060 | 052 | 090 | 001 | 001 | 001 | 160 | 010 | 280 | 260 | 170 |
| 216 | 270 | 170 | 025 | 040 | 001 | 001 | 001 | 160 | 010 | 290 | 070 | 330 |
| 217 | 250 | 185 | 031 | 025 | 001 | 001 | 001 | 120 | 001 | 260 | 080 | 330 |
| 218 | 260 | 185 | 030 | 015 | 001 | 001 | 001 | 270 | 080 | 480 | 010 | 330 |
| 219 | 270 | 185 | 032 | 010 | 005 | 001 | 001 | 180 | 040 | 450 | 020 | 220 |
| 220 | 325 | 045 | 053 | 005 | 020 | 001 | 001 | 600 | 080 | 660 | 020 | 250 |
| 221 | 315 | 090 | 047 | 005 | 020 | 001 | 001 | 410 | 200 | 600 | 060 | 260 |
| 222 | 335 | 100 | 047 | 010 | 040 | 001 | 001 | 360 | 080 | 590 | 110 | 170 |
| 223 | 310 | 010 | 049 | 005 | 080 | 018 | 001 | 640 | 240 | 630 | 060 | 190 |
| 224 | 410 | 001 | 049 | 001 | 075 | 032 | 001 | 760 | 440 | 800 | 001 | 001 |
| 225 | 360 | 001 | 048 | 001 | 080 | 055 | 001 | 770 | 260 | 770 | 010 | 010 |

Table 3a (Cont'd.)

| ID | $Z_1$ | $Z_2$ | $Z_3$ | $Z_4$ | $Z_5$ | $Z_6$ | $Z_7$ | $Z_8$ | $Z_9$ | $Z_{10}$ | $Z_{11}$ | $Z_{12}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 226 | 310 | 015 | 051 | 001 | 105 | 036 | 001 | 660 | 380 | 640 | 001 | 010 |
| 227 | 420 | 005 | 049 | 001 | 095 | 056 | 001 | 620 | 520 | 680 | 001 | 001 |
| 228 | 415 | 020 | 049 | 005 | 025 | 036 | 001 | 370 | 220 | 340 | 001 | 001 |
| 229 | 420 | 005 | 041 | 001 | 070 | 060 | 001 | 630 | 510 | 580 | 001 | 001 |
| 230 | 450 | 005 | 040 | 001 | 090 | 070 | 001 | 690 | 570 | 630 | 001 | 001 |
| 231 | 395 | 001 | 025 | 015 | 100 | 071 | 001 | 580 | 530 | 560 | 001 | 010 |
| 232 | 380 | 010 | 027 | 025 | 035 | 039 | 001 | 350 | 320 | 400 | 001 | 270 |
| 233 | 430 | 010 | 025 | 030 | 030 | 025 | 001 | 340 | 340 | 360 | 001 | 200 |
| 234 | 410 | 075 | 022 | 010 | 005 | 015 | 001 | 170 | 170 | 170 | 001 | 060 |
| 235 | 520 | 055 | 024 | 040 | 005 | 001 | 001 | 210 | 190 | 190 | 001 | 180 |
| 236 | 385 | 135 | 018 | 010 | 005 | 008 | 001 | 140 | 200 | 260 | 001 | 020 |
| 237 | 535 | 065 | 010 | 020 | 001 | 001 | 001 | 110 | 230 | 270 | 001 | 070 |
| 238 | 550 | 095 | 001 | 010 | 001 | 001 | 001 | 050 | 230 | 270 | 001 | 030 |
| 239 | 510 | 100 | 001 | 001 | 001 | 001 | 001 | 190 | 150 | 230 | 001 | 110 |
| 240 | 510 | 095 | 001 | 040 | 001 | 001 | 001 | 140 | 100 | 150 | 001 | 040 |
| 241 | 385 | 180 | 010 | 001 | 001 | 001 | 001 | 050 | 050 | 300 | 001 | 050 |
| 242 | 505 | 125 | 001 | 001 | 001 | 001 | 001 | 001 | 200 | 130 | 001 | 030 |
| 243 | 470 | 090 | 001 | 020 | 001 | 001 | 001 | 160 | 300 | 380 | 001 | 060 |
| 244 | 465 | 110 | 001 | 035 | 001 | 001 | 001 | 260 | 440 | 500 | 001 | 060 |
| 245 | 400 | 140 | 001 | 015 | 001 | 023 | 001 | 330 | 400 | 390 | 001 | 040 |
| 246 | 415 | 105 | 015 | 023 | 040 | 032 | 001 | 220 | 190 | 270 | 001 | 010 |
| 247 | 435 | 075 | 010 | 015 | 001 | 069 | 001 | 370 | 360 | 500 | 001 | 010 |
| 248 | 370 | 145 | 010 | 010 | 005 | 012 | 040 | 130 | 080 | 330 | 001 | 030 |
| 249 | 380 | 210 | 001 | 001 | 001 | 001 | 020 | 070 | 001 | 050 | 001 | 030 |
| 250 | 430 | 065 | 001 | 005 | 020 | 001 | 075 | 130 | 070 | 300 | 001 | 020 |
| 251 | 420 | 080 | 030 | 001 | 005 | 026 | 001 | 050 | 100 | 350 | 001 | 050 |
| 252 | 425 | 060 | 035 | 005 | 001 | 001 | 030 | 100 | 010 | 340 | 001 | 010 |
| m | 250 | 001 | 001 | 001 | 001 | 001 | 001 | 001 | 001 | 050 | 001 | 001 |
| M | 520 | 210 | 065 | 090 | 105 | 071 | 075 | 770 | 570 | 800 | 270 | 570 |

Table 3b

Details on Data for Example 2

| Feature Variable | Set A | | Set B | | Set C | | Set D | | Set E | | Set F | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Range | Data | Range | Data | Range | Data | Range | Data | Range | Data | Range | Data |
| $x_1$ | .8 | | .8 | | .8 | | .8 | | .8 | | .8 | |
| $x_2$ | (.2,.8) | $z_1$ | (.2,.8) | $z_1$ | (.2,.8) | $z_8$ | .5 | | .5 | | .5 | |
| $x_3$ | .9 | | .9 | | .9 | | .9 | | .9 | | .9 | |
| $x_4$ | .9 | | .9 | | .9 | | 1.4 | | .8 | | 1.4 | |
| $x_5$ | (1,2) | $z_2$ | (1,2) | $z_2$ | (1,2) | $z_9$ | .8 | | 1.4 | | .8 | |
| $x_6$ | (.1,.5) | $z_3$ | (.1,.5) | $z_3$ | .5 | | (.1,.5) | $z_3$ | .5 | | .5 | |
| $x_7$ | (.2,.8) | $z_4$ | (.2,.8) | $z_4$ | (.1,.5) | $z_{10}$ | (.2,.8) | $z_4$ | (.2,.8) | $z_4$ | (.2,.8) | $z_{10}$ |
| $x_8$ | (.3,2.0) | $z_5$ | (.3,2.0) | $z_5$ | (.2,.8) | $z_{11}$ | (.3,2.0) | $z_5$ | 1.0 | | 1.0 | |
| $x_9$ | (.2,.9) | $z_6$ | (.2,.9) | $z_6$ | (.3,2.0) | $z_{12}$ | .5 | | .5 | | .5 | |
| $x_{10}$ | (.1,.8) | $z_8$ | .5 | | .5 | | (.1,.8) | $z_8$ | (.1,.8) | $z_1$ | (.1,.8) | $z_8$ |
| $x_{11}$ | (.2,.8) | $z_9$ | .5 | | .5 | | (.2,.8) | $z_9$ | (.2,.8) | $z_2$ | (.2,.8) | $z_9$ |
| $x_{12}$ | (.5,.8) | $z_{10}$ | .5 | | .5 | | (.3,.8) | $z_{10}$ | (.3,.8) | $z_3$ | (.3,.8) | $z_{11}$ |
| $x_{13}$ | (.4,.8) | $z_{11}$ | .6 | | .6 | | (.4,.8) | $z_{11}$ | (.4,.8) | $z_5$ | (.4,.8) | $z_{12}$ |
| $x_{14}$ | (.2,.8) | $z_{12}$ | .5 | | .5 | | (.4,.8) | $z_{12}$ | (.4,.8) | $z_6$ | .6 | |
| $x_{15}$ | .5 | | .5 | | .5 | | (.1,.9) | $z_1$ | .5 | | .5 | |
| $x_{16}$ | (.3,.6) | $z_7$ | (.3,.6) | $z_7$ | .4 | | (.2,.8) | $z_7$ | (.2,.8) | $z_7$ | .4 | |
| $x_{17}$ | .5 | | .5 | | .5 | | (.2,.8) | $z_2$ | .5 | | .5 | |
| $x_{18}$ | .5 | | .5 | | .5 | | (.2,.8) | $z_6$ | .5 | | .5 | |

## Table 3c

### Dictionary for Specimens in Figures 3A-3F

| ID(1) | Integer | ID(1) | Integer | ID(1) | Integer | ID(1) | Integer |
|---|---|---|---|---|---|---|---|
| 200.000 | 1532932388 | 227.000 | 1290947939 | 205.000 | 10617 | 218.000 | 20871 |
| 201.000 | 633332057 | 228.000 | 934725132 | 229.000 | 10824 | 222.000 | 20873 |
| 202.000 | 1482927467 | 229.000 | 1082415719 | 231.000 | 11352 | 204.000 | 21172 |
| 203.000 | 2014214434 | 230.000 | 827015496 | 238.000 | 11548 | 235.000 | 21684 |
| 204.000 | 2117264577 | 231.000 | 1135277888 | 208.000 | 11980 | 232.000 | 22326 |
| 205.000 | 1061714849 | 232.000 | 223260021 | 209.000 | 12003 | 219.000 | 23251 |
| 206.000 | 799844220 | 233.000 | 677241638 | 242.000 | 12375 | 225.000 | 24447 |
| 207.000 | 1881658967 | 234.000 | 736880766 | 249.000 | 12524 | 246.000 | 26279 |
| 208.000 | 1198072647 | 235.000 | 216841913 | 212.000 | 12894 | 244.000 | 29127 |
| 209.000 | 1200303857 | 236.000 | 182282832 | 227.000 | 12909 | 210.000 | 45544 |
| 210.000 | 45544681 | 237.000 | 1315876802 | 223.000 | 13115 | 250.000 | 46205 |
| 211.000 | 965275235 | 238.000 | 1154814408 | 237.000 | 13158 | 251.000 | 51382 |
| 212.000 | 1289405207 | 239.000 | 8553670 | 245.000 | 13385 | 248.000 | 61785 |
| 213.000 | 775832172 | 240.000 | 2027610988 | 202.000 | 14829 | 216.000 | 62992 |
| 214.000 | 2038093867 | 241.000 | 1787364720 | 200.000 | 15329 | 201.000 | 63333 |
| 215.000 | 1879453019 | 242.000 | 1237594804 | 224.000 | 16008 | 233.000 | 67724 |
| 216.000 | 629926610 | 243.000 | 1876749633 | 247.000 | 16379 | 226.000 | 71100 |
| 217.000 | 82154560 | 244.000 | 291274695 | 220.000 | 16720 | 234.000 | 73688 |
| 218.000 | 2087188546 | 245.000 | 1338567352 | 221.000 | 16795 | 252.000 | 75608 |
| 219.000 | 232518877 | 246.000 | 262799092 | 241.000 | 17873 | 213.000 | 77583 |
| 220.000 | 1672011846 | 247.000 | 1637961012 | 236.000 | 18228 | 206.000 | 79984 |
| 221.000 | 1679574727 | 248.000 | 617857791 | 243.000 | 18767 | 217.000 | 82154 |
| 222.000 | 2087380521 | 249.000 | 1252460092 | 215.000 | 18794 | 230.000 | 82701 |
| 223.000 | 1311559055 | 250.000 | 462058350 | 207.000 | 18816 | 239.000 | 85536 |
| 224.000 | 1600884577 | 251.000 | 513820898 | 203.000 | 20142 | 228.000 | 93472 |
| 225.000 | 244472376 | 252.000 | 756088099 | 240.000 | 20276 | 211.000 | 96527 |
| 226.000 | 711006721 | | | 214.000 | 20380 | | |

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